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Experimental and modeling study of styrene oxidation in spherical reactor and shock tube

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ABSTRACT

The oxidation of styrene, one of the main stable intermediates from the oxidation of large alkylated aromatic hydrocarbons, has been investigated in the present work both experimentally and numerically. Experiments were performed using two complementary techniques, spherical bomb for laminar flame speed studies and shock tube for ignition delay time measurements. In particular, the laminar flame speeds of styrene/air mixtures were measured at three different initial temperatures (342 K, 373 K, and 405 K), over a wide range of equivalence ratios (0.75–1.45), for an initial pressure of 100 kPa. In addition, the autoignition of styrene/O_2 mixtures in argon bath gas ($\phi = 0.5$, 1.0, and 1.5) was investigated over a wide range of temperatures (1390–1990 K), at highly diluted conditions (94.3%–99% argon), and for pressures between 110 and 200 kPa. A detailed chemical kinetic model, based on the toluene chemistry by Metcalfe et al. (2011), was developed and validated against the newly obtained experimental results and the flow reactor data available in the literature (Litzinger et al., 1986). Sensitivity and rate of production analyses were performed and showed that, at the conditions studied herein for the flame speed investigation, the main fuel consumption pathways include the reaction of the fuel with H atoms to form phenyl radical and ethylene or benzene and vinyl radical, with O to form benzyl radical + HCO, and the H-abstraction reactions on both the vinyl moiety and the ring. On the other hand, the analyses performed at the high-temperature, highly-diluted conditions encountered in the shock tube study highlighted the importance of the fuel thermal decomposition steps for the simulation of the ignition delay time measurements. The model was also tested against the low-pressure flame and jet stirred reactor data by Yuan et al. (2015). The results highlight the need for future modifications in the benzene chemistry by Metcalfe et al. and inclusion of pressure dependent rate parameters in order to improve the prediction capabilities of the model.

1. Introduction

Styrene is a fundamental intermediate in the high-temperature oxidation of alkylated aromatic compounds and consequently of commercial fuels which contain a substantial portion of aromatic components. For example, in their study on the oxidation of ethylbenzene performed using the Princeton atmospheric flow reactor, Litzinger et al. [1] measured large amounts of styrene produced from the dehydrogenization of the fuel. In order to facilitate the kinetic analysis, an experimental set was also conducted with styrene as the fuel molecule, at fuel-lean conditions ($\phi = 0.56$), and at an initial temperature of around 1060 K. Subsequent studies performed by the same research group on larger alkylated single-ring aromatics, such as *n*-propylbenzene [2,3] and *n*-butylbenzene [4], suggested once again the importance of the pathways to styrene. Other experimental investigations on these large alkylated compounds highlighted styrene as one of the key aromatic intermediates, including the species measurements in jet-stirred reactors by Dagaut et al. [5], by Diévart and Dagaut [6] and by Husson et al. [7], and in a high-pressure shock tube by Gudiyella and Brezinsky [8]. Indeed, the accuracy in the description of the chemistry involved in the styrene oxidation does not only influence the ability to predict the oxidation of large aromatics and fuel mixtures, but also the formation of polycyclic aromatic hydrocarbons and finally soot. In fact, the oxidation of styrene leads to the formation of several intermediates, such as phenylacetylene, 2-phenylvinyl (C_6H_5CHCH), and 2-vinylphenyl ($C_6H_4C_2H_3$), which constitute essential building blocks for the growth to the second-ring structure [9–11].

Despite the central role of styrene in the oxidation of aromatic compounds, few investigations have been specifically focused on its chemistry. As mentioned above, the flow reactor study by Litzinger et al. [1] provides species profiles for the main in-

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termediates and products of the styrene oxidation. In a recent investigation, Yuan et al. [12] measured species profiles both in laminar premixed flames of styrene at low pressure (0.0395 atm, ϕ = 0.75, 1.00, and 1.70) using synchrotron vacuum ultraviolet photoionization mass spectrometry and in a jet-stirred reactor for styrene/benzene mixtures at atmospheric pressure using gas chromatography. The authors also developed a detailed chemical kinetic model to simulate the species profiles previously obtained. Other studies on styrene include the works by Müller-Markgraf and Troe [13] and by Grela et al. [14] which performed kinetic experiments to derive the reaction rate constant of the styrene dissociation to benzene and vinylidene. In the present investigation, new experimental results on the laminar flame speeds of styrene/air mixtures have been obtained using a heated spherical reactor. Ignition delay time measurements in a glass shock tube were also performed as a complement to the flame speed study. Finally, a chemical kinetic model was developed to simulate the newly obtained experimental results as a base for the development of kinetic models for larger alkylated compounds and fuel surrogates.

2. Experimental and modeling techniques

2.1. Spherical bomb

Details of the experimental apparatus have been reported in a recent publication [15] and only the main features are described below. The spherical bomb consists of a spherical stainless steel vessel of 56-L equipped with 4 quartz windows (100 mm optical diameter) and homogeneously heated with a maximum uncertainty of ±1 K. In the present study, the initial temperature was varied (342 ±1 K, 373 ±1 K, and 405 ±1 K) while the initial pressure in the spherical bomb was fixed to 100 kPa for all the experimental sets presented. The liquid fuels and the gases were introduced directly in the vessel using the partial pressure method. The pressures were measured using capacitive manometers (MKS Baratron, Type 631) of different full scales according to the desired pressure range (100 and 1000 Torr). Based on the precision of the capacitive manometers (0.5% of the reading), the partial pressure of each component could be estimated with an accuracy of ~0.5%, thus the maximum error on the equivalence ratio is around 1%. This value is a conservative estimate of the real error since the manometers are periodically calibrated against a reference capacitive manometer (for the 0–100 Torr) or against a mercury manometer (0–1000 Torr) with small deviations between the measurements corrected by the calibration curve and the reference values (lower than 0.5%).

The combustion process, initiated at the center of the sphere by an electric spark obtained using tungsten electrodes, was monitored using two different diagnostics: pressure measurements using a piezo-electric pressure transducer (Kistler 601A coupled to a Kistler Type 5011B Charge Amplifier) and recording of the flame. The visualization of the flame was obtained via a Schlieren diagnostic coupled with a high speed camera (Phantom v1610). The frame rate was fixed to 25,000 i/s. An average increase of around 0.75% was observed in the pressure traces between the TTL signal (produced by a digital delay/pulse generator (Stanford Research System INC) at the spark and used for the synchronization of the camera and the oscilloscope) and the time corresponding to the last image which could be observed and measured. Examples of flame images and pressure trace are presented in Fig. 1.

2.2. Laminar flame speed determination and related uncertainties

Automated software was used to analyze the images (i.e., in Fig. 1) and to obtain the radius of the flame (Canny method) as a function of time from which the unstretched laminar flame speed of the unburned gases is derived. In particular, the following relation by Ronney and Sivashinsky [16], as subsequently modified by Kelley and Law [17], was solved iteratively to obtain both the adiabatic unstretched gas speed of the burned gases relative to the flame, S_b^0, and the Markstein length, L_b,

$$\left(\frac{S_b}{S_b^0}\right)^2 \cdot \ln\left(\frac{S_b}{S_b^0}\right)^2 = -\frac{2L_b \cdot \kappa}{S_b^0} \tag{1}$$

where κ is the stretch rate. The value S_b^0 and L_b are determined using the following steps: (i) a couple of values of S_b^0 and L_b are chosen to initialize the optimization process, (ii) the non-linear equation is solved to get the radii predicted by the equation based on the values of the chosen S_b^0 and L_b, (iii) these radii are compared to the experimental ones, and if they differ, a new set of S_b^0 and L_b is chosen. A minimization process is then applied in order to obtain the best correspondence between the experimental radii and the ones derived from the nonlinear equation. A Matlab processing is used to perform these optimization steps. The minimum radius used in the data processing was around 10 mm, while the maximum radius between 30 and 35 mm. This range was chosen to guarantee the optimal quality of the flame images and accuracy of the consequent experimental profiles.

The unstretched laminar flame speed of the unburned gases, S_u^0, was derived from the continuity equation. For a flame propagating at constant pressure conditions, as in the present experiments, $S_u^0 = S_b^0/\sigma$, where $\sigma = \rho_u/\rho_b$ and ρ_u, ρ_b are the unburned and burned densities of the mixture, respectively. The densities of unburned and burned gases, ρ_u and ρ_b, were calculated using COSILAB© [18] with the Equilibrium code. The maximum estimated error in the derivation of the laminar flame speed is around ±1 cm/s which derives from the estimated uncertainty in the determination of the radii (±1 pixel).

Examples of unstretched spatial laminar burning speed extractions using the non-linear method (NQ based on the notation in Ref. [19]) are shown in Fig. 2. The solution of the linear relation between burning speed and stretch rate ($S_b = S_b^0 - L_b \cdot \kappa$, [20,21], named LS) is also reported for comparison. The symbols in Fig. 2 correspond to the spatial laminar burning speed derived from the measured radius versus time: $S_b(R_i) = \frac{R_{i+1}-R_{i-1}}{t_{i+1}-t_{i-1}}$. The radius is not fitted using an expression of dR_f/dt and the plotted symbols corresponding to S_b are presented for display purposes only. The solid line corresponds to the non-linear solution and the dashed line to the linear solution. While at fuel rich conditions the two unstretched laminar burning speeds appear to be nearly identical, the solutions diverge moving toward lean mixtures. These results highlight the influence of the extrapolation method based on the current (R_f, t) datasets, and they do not provide any information on the global uncertainty in the laminar flame speed measurements. Nevertheless, in order to better assess such influence, additional methods were implemented including the two models investigated by Chen [22] (NM-I and NM-II) and the non-linear model in expansion form (NE based on the notation in Ref. [19]). The comparison between the results is reported in Fig. 3 for the case of initial temperature equal to 342±1 K, although similar results were obtained for the other datasets.

Excluding the linear model which is not a realistic model to describe correctly the propagation of expanding flames over the entire range of equivalence ratios studied herein, the use of NM-I (solid line), NM-II (dash-dot line), and NE (dash line) leads to results which are within the experimental uncertainties of the measurements obtained solving Eq. 1, represented by the symbols in Fig. 3. In particular, all the methods converge at fuel rich conditions, including the linear model as already mentioned concerning Fig. 2, while they diverge at lean conditions. In particular, the

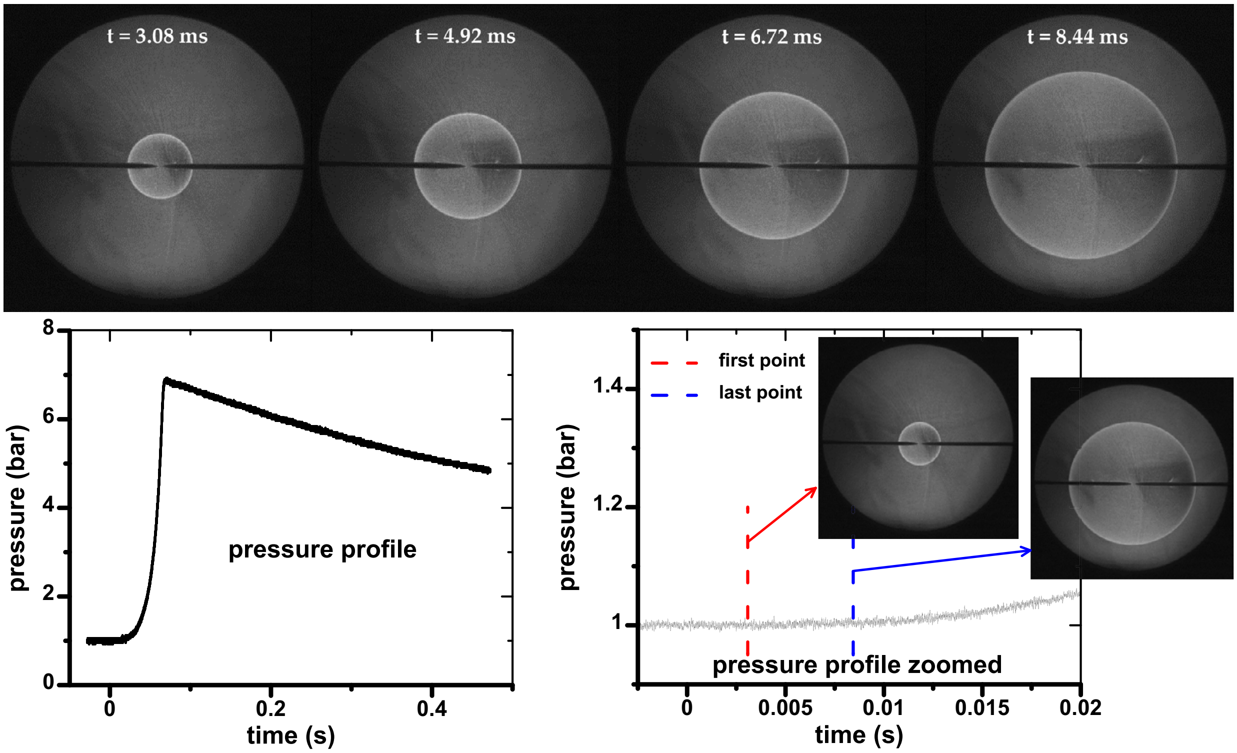

Fig. 1. Styrene flame propagation, $\phi = 1.13$, $T_{in} = 405$ K, $P_{in} = 100$ kPa. The corresponding pressure profile is also reported.

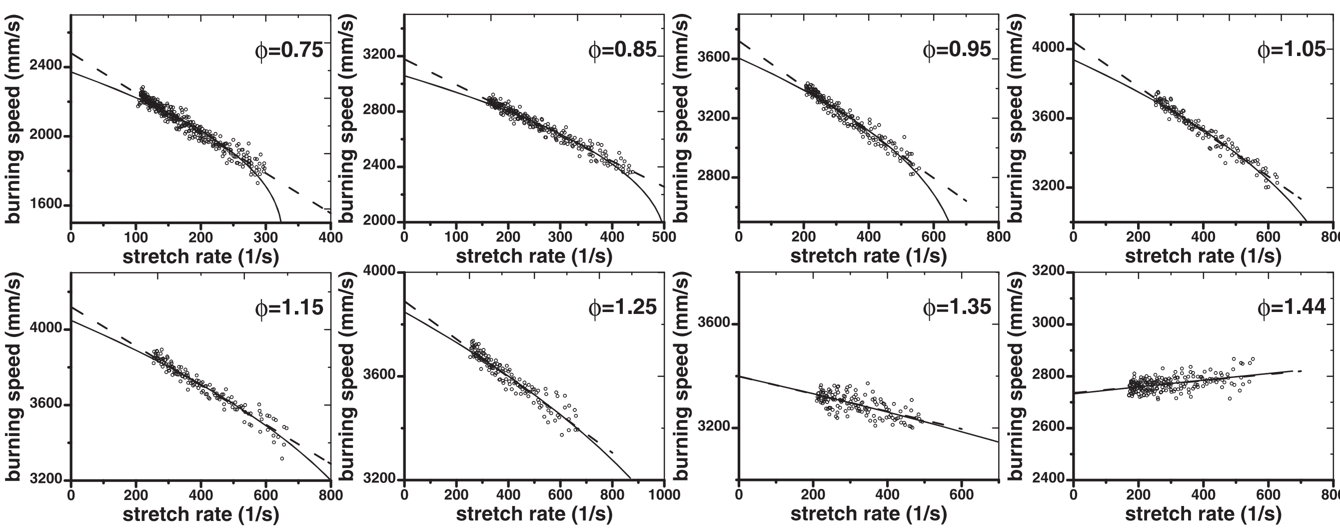

Fig. 2. Evolution of the spatial laminar burning speed versus stretch rate for styrene/air mixtures initially at 373 K and 100 kPa.

NM-II model provides results which are very similar to the solution of Eq. 1. This is expected since in reality the expression derived for NM-II can be obtained by simply substituting the expression of the stretch rate into Eq. 1. On the other hand, NM-II can be solved with a least-square method fitting the data of $\ln(S_L)$ as function of $1/(S_L{*}r)$, while in the present work the (S_b^0, L_b) solution is obtained through the minimization process described above which is based on the actual measured radii. Thus, the comparison between the two solutions provides a confirmation for the validity of our mathematical approach. Compared to the results obtained solving Eq. 1, NM-I and NE results are slightly higher (the two corresponding curves in Fig. 3 are almost overlying). The maximum discrepancy is observed between the NQ and the NE model, with an average and maximum differences of 0.9% and 2.0%, respectively. The good agreement between the different extrapolation methods is mainly due to two factors, the dimensions of the flame radii used to derive the laminar flame speed (the quasi steady-state propagation applies over a large domain without interferences from the walls) and the large number of experimental points (radius vs time) used in the correlations.

The uncertainty in the extrapolation process can also be verified by analyzing the product between the Markstein number and the Karlovitz number as suggested by Wu et al. [19], although such verification method will require further validations over a wide range of fuels (Wu et al. [19] considered hydrogen and n-heptane in their analysis). Figure 4 contains the results of such analysis which was performed considering the linear Markstein number and the Karlovitz number at the middle radius (as in [19]) for

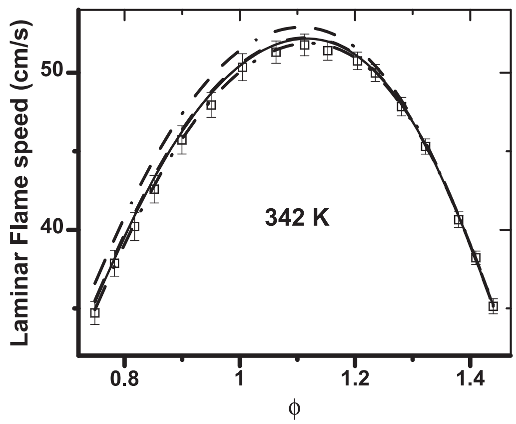

Fig. 3. Comparison between different extrapolation methods at T_{in} = 342 K and P_{in} = 100 kPa. The symbols represent the laminar flame speed measurements obtained with the non-linear method NQ with the relative estimated errors. Other models: —– NM-I, – • – NM-II, – – NE, – – • – – LS.

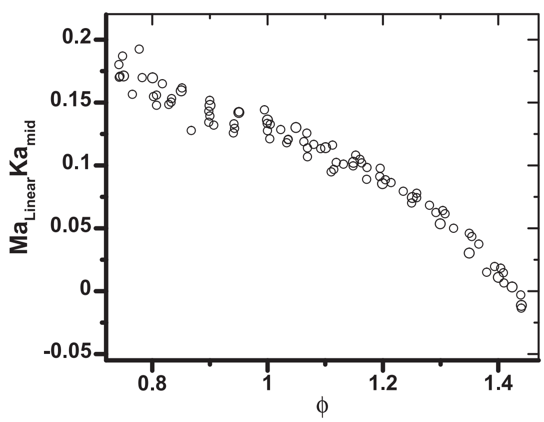

Fig. 4. (Ma_{Linear} × Ka_{mid}) as function of the equivalence ratio. T_{in} = 342 K, 373 K, 405 K; P_{in} = 100 kPa.

all the experiments performed. Over almost the entire range of equivalence ratios considered the product is below 0.15, condition for which negligible extrapolation uncertainty is expected. Only at very lean conditions (ϕ between 0.75 and 0.80, approximately) the laminar flame speed results may be slightly less accurate, in agreement with the fact that in Fig. 3 the largest discrepancies occur in this range of equivalence ratios.

Another aspect which needs to be considered in order to assess the accuracy of the experimental results is the effects of radiative heat losses. As for the case of pentanol/air mixtures investigated in [15], the flame speed profiles presented in Fig. 2 do not show any bending due to radiation. Such bending is typical of flames which are strongly affected by radiation losses, as investigated by Jayachandran et al. [23] using DNS simulations of CH_4/air flames. This observation suggests that the flames studied here are not strongly affected by this phenomenon, in agreement with several studies showing how the radiative heat losses play an important role only when the flame speed is low (i.e., at high pressure or near the flammability limits) or when the mixtures are diluted with CO_2 or H_2O_{vap} [24-26]. The radiation-correction terms calculated based on the equation proposed by Yu et al. [27] are shown in Fig. 5. For the three initial temperature conditions of 342 K, 373 K, and 405 K the correction is on average equal to 0.55 cm/s, 0.59 cm/s, and 0.63 cm/s, respectively. These corrections will be included in the experimental errors in Figs. 7 and 8.

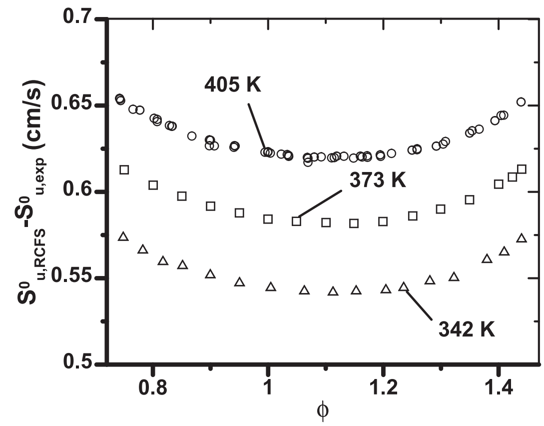

Fig. 5. Radiation correction estimated according to Yu et al. [27]. P_{in} = 100 kPa.

2.3. Shock tube

The shock tube is composed of a 9 m long unheated glass driven section and a 2 m long stainless steel driver section with constant internal diameter (50 mm) [28]. A metallic knife attached to the extremity of a rod is mechanically displaced inside the driver section to generate the shock waves by rupture of the terphane diaphragm located between the driven and the driver sections. The pressure profiles from four CHIMIE METAL A25L05B sensors provided the measurement of the incident shock wave velocities. Such velocities were subsequently used together with the initial conditions to calculate the pressure and temperature conditions behind the reflected shock waves by solving the conservation equations. The method used for the calculations assumes thermal equilibrium and no reaction before ignition as well as variable γ. The estimated errors in the temperature and pressure calculations are respectively 1% and 1.5%. In addition, spectrophotometric measurements were performed using a HAMAMATSU R928 photomultiplier with a monochromator centered at 306 nm for detection of the OH∗ radicals emissions. In the present work, the ignition delay time was defined as the time between the arrival of the reflected shock wave at the measurement location and the time when the emission profile reaches 50% of its maximum value. The uncertainty in the ignition delay time measurement is estimated between 2% for the longest ignition delay times up to 14% at high temperature conditions. Such uncertainty is mainly due to the shape of the OH∗ emission profile and the consequent error in the identification of the ignition point. As a percentage of the total ignition delay time, such error will be higher at shorter ignition delay times, as mentioned above. Experiments were also conducted in inert argon gas using a PCB 102A06 pressure transducer mounted at the end-wall perpendicularly to the flow to monitor the pressure in the tube during the observation time. Typical pressure and OH∗ profiles are provided in Fig. 6. Note that the increase in the side-wall pressure profile on the left graph is due only to the heat transfer to the CHIMIE METAL sensor which is very sensitive to temperature changes. In fact, the corresponding end-wall pressure trace obtained with argon driven gas (right graph) does not present any increment.

2.4. Reagents

Commercial styrene contains a certain amount of stabilizer (*p*-tert-butylcatechol in the present case) which is added to the fuel in order to minimize the polymerization process during storage. Two different styrene samples were considered in the present study in order to assess the influence of the impurities

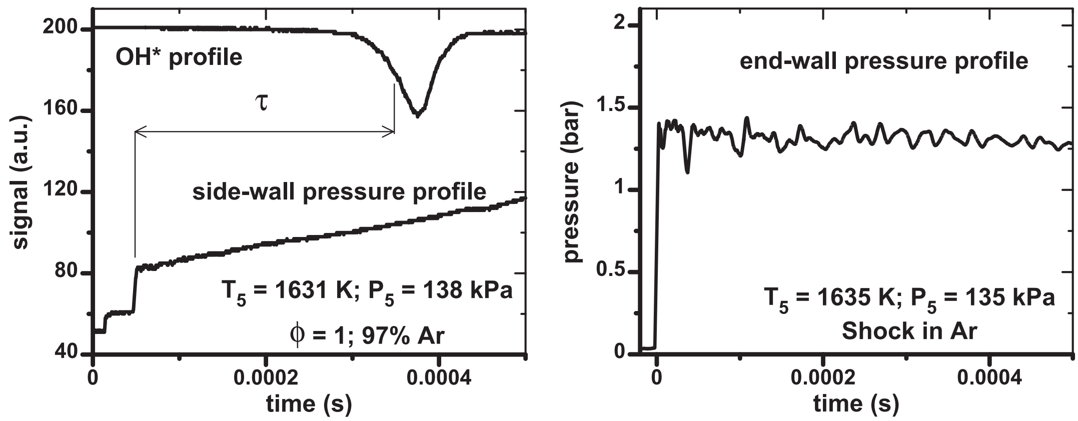

Fig. 6. Typical OH* and pressure profiles obtained with a reactive mixture (left) and a non-reactive one (right).

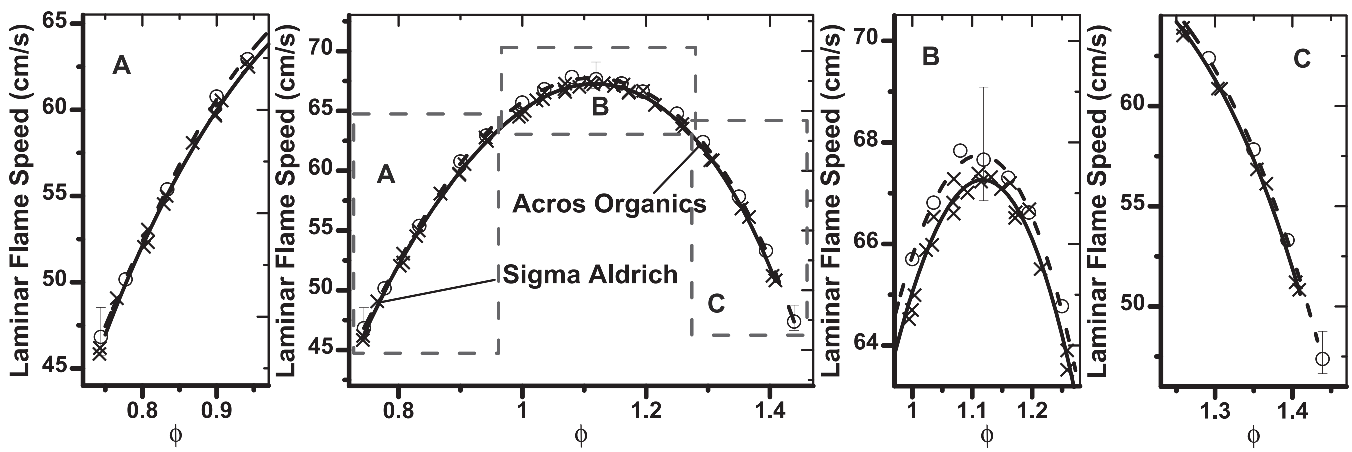

Fig. 7. Laminar flame speeds of styrene/air mixtures. T_{in} = 405 K; P_{in} = 100 kPa. Styrene by: crosses, —– Sigma Aldrich; circles, – – Acros Organics. Solid lines (Sigma Aldrich) and dashed lines (Acros Organics) represent the polynomial fits.

and stabilizer mole fractions on the experimental results. The first styrene sample by Sigma Aldrich (ReagentPlus, >99% purity) contains relatively high amounts of the stabilizer (50 ppm), while the second one by Acros Organics (>99.5% purity) only 10 to 20 ppm of *p*-tert-butylcatechol. The comparison between the two fuels has been conducted based on the flame speeds measured at an initial temperature of 405 K (Fig. 7). For clarity of presentation, the error bars have been included only for few representative experiments. The two experimental sets show a slight shift with the styrene from Acros Organics, characterized by higher purity and lower stabilizer concentration, having higher flame speeds over almost the entire equivalence ratio range. The polynomial fits presented in Fig. 7 can be used in order to further estimate the difference between the experimental sets. In fact, the discrepancy between fits and measurements is always less than 0.5 cm/s, specifically less than 0.3 cm/s for around 90% of the data. On average the two polynomial fits differ by around 0.5 cm/s in the equivalence ratio range between 0.75 and 1.40, below the experimental uncertainty on the single measurement. Thus, the difference in the amounts of impurities and inhibitor does not influence the flame speed results in a measurable way. For the remaining experiments conducted with the spherical bomb at different temperatures the Acros Organics styrene was used while laboratory dry air consisted of 0.209 O_2 + 0.791 N_2. The styrene sample from Sigma-Aldrich was utilized to perform the experiments on the ignition delay times.

As mentioned above, styrene isomerizes quite rapidly, with a rate of 0.1% per hour at 60°C and 2% per hour at 100°C (data for purified styrene [29]). In order to reduce the experimental uncertainties, the time employed to run the flame speed experiments was minimized and the experimental procedure was maintained consistent during the entire study. Of course, the measurements performed at 342 K will be almost unaffected by the polymerization process, while at higher temperatures a larger uncertainty in the absolute flame speed values should be considered. On the other hand, no evident sign of the polymerization process, such as for example condensation of solid particles on the glass windows, could be observed even at 405 K. The experiments conducted with the shock tube at room temperature should not be affected by any degradation of the fuel.

2.5. Chemical kinetic model

The chemical kinetic model developed here is based on the extensive work by Metcalfe et al. [30] on the pyrolysis and oxidation of toluene. Since the model was validated against a variety of experimental results, including species profiles (in flow reactors, jet-stirred reactors, and shock tubes), ignition delay times, and flame speeds for toluene, benzene, and cyclopentadiene, no modifications were made on the aromatic chemistry up to toluene, including the C0-C4 submodel and the chemistry of benzene and the phenyl radical, cyclopentadiene, phenol and the phenoxy radical. More information on this part of the model can be obtained directly from Ref. [30]. The model proposed by Metcalfe et al. [30] also contains a sub-set of reactions for styrene and ethylbenzene which was replaced to better fit the new experimental data. In particular, the reaction rate parameters for the thermal decomposition of styrene to benzene and vinylidene were taken from the work by Yuan et al. [12] who divided by a factor of 5 the high-pressure limit rate

Table 1
Activation energies of H-abstraction reactions by H, OH, and O for propene [34], and corresponding corrections for styrene (kcal/mol).

	From CH	From CH_2	Correction
H	9790	12,281	2491
OH	1451	2782	1331
O	7631	8962	1331

constant by Grela et al. [14]. The H-abstraction reactions by radicals on the vinyl moiety were estimated based on the corresponding reactions for ethylene (for the terminal carbon) and propene. In particular, the parameters for the abstraction on the terminal carbon by H atoms were taken from the work by Knyazev et al. [31], by OH atoms from the JetSurF 2.0 model [32], and by O atoms from the work by Mahmud et al. [33], with the pre-exponential factors divided by 2 to account for the different number of hydrogen atoms available. These rate constants were used as a reference to derive the corresponding parameters for the H-abstraction from the carbon adjacent to the ring, with the pre-exponential factors divided by 2 (only one hydrogen can be removed compared to the two hydrogens on the terminal carbon) and the activation energies corrected based on the reactions of propene [34]. The corrections on E_a are presented in Table 1 together with the activation energies of the propene reactions. The reaction rate parameters of the H-abstraction reactions on the ring were based on the parameters for benzene and toluene available in the literature, corrected to account for the multiplicity. In particular, the rate constant for the abstraction by OH radicals from the CRECK model was used [35] and by H and O atoms from the original model by Metcalfe et al. [30]. The decomposition and isomerization reactions of the primary radicals of the styrene oxidation, i.e., 2-phenylvinyl, 1-phenylvinyl ($C_6H_5CCH_2$) and 2-vinylphenyl, were taken from the work by Tokmakov and Lin [36], while the oxidation steps were estimated based on the reactions of the vinyl radical. The pre-exponential factor of the reaction between phenyl and ethylene forming styrene and H (and consequently the one of the reverse reaction) was increased by a factor of 1.5 compared to the calculations by Tokmakov and Lin [37] since the reaction rate constant measured by Fahr and Stein [38] between 1000 and 1330 K is, on average, higher by a factor of 1.5 compared to the value in [37]. The parameters of the ipso-additions by H and OH radicals on styrene were estimated based on the corresponding parameters for toluene ([30] and [39]), while the global reactions $C_6H_5C_2H_3$+OH to form C_6H_5CHO+CH_3 or $C_6H_5CH_2$+CH_2O and $C_6H_5C_2H_3$+O to form $C_6H_5CH_2$+HCO were taken from Narayanaswamy et al. [40] and from the reaction of C_2H_4+O forming CH_3+HCO [32], respectively. The oxidation reactions involving phenylacetylene which will be relevant to the subsequent discussions include the reactions with OH to form the phenyl radical + ethenone (CH_2CO) from Yuan et al. [12], the benzyl radical + CO from the CRECK model [35], or benzene + HCCO from Narayanaswamy et al. [40], the reactions with O to form the phenyl radical + HCCO (in analogy to the reaction C_2H_2+O=HCCO+H, [32]) or phenylcarbene (C_6H_5CH) + CO (rate constant parameters from Eichholtz et al. [41]) and the subsequent oxidation reactions of phenylcarbene from the MIT atmospheric pressure soot model [42]. The abstraction reactions from the phenylacetylene ring were based on the corresponding reactions for toluene. Finally, the sub-model for ethylbenzene oxidation was based on the work by Nakamura et al. [43], while the reactions involving indene and naphthalene were taken from the work by Narayanaswamy et al. [40]. The final model is composed of 2016 reactions and 336 species, including 1843 reactions from the model of Metcalfe et al. [30] and the new styrene sub-mechanism composed of 173 reactions.

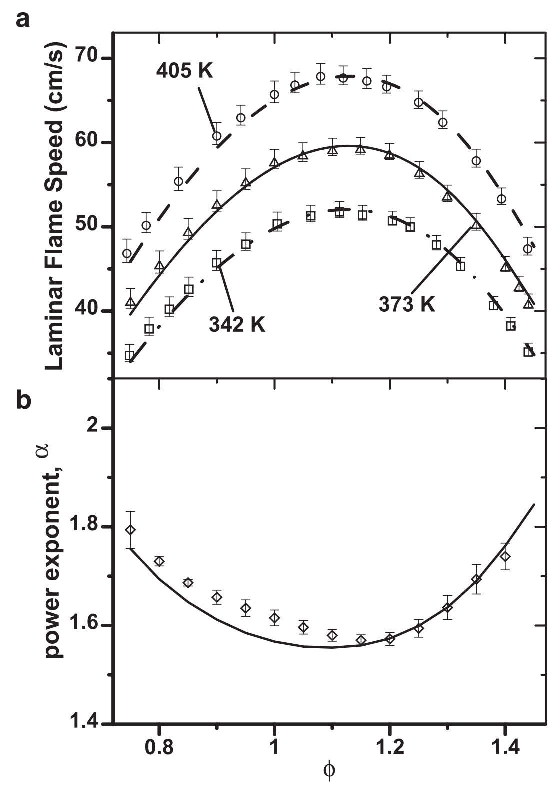

Fig. 8. Laminar flame speeds of styrene/air mixtures, P_{in} = 100 kPa. Initial temperature equal to: (a) circles, – – 405 K, triangles, —– 373 K, squares, – • – 342 K; (b) experimental power exponent α (symbols). Lines represent modeling results.

All the simulations have been performed using Cosilab software, version 3.3.2 [18]. For the flame speed calculations the one-dimensional freely-propagating flame system was solved, with GRAD and CURV values equal to 10^{-5} and a spatial domain from −1 to 5 cm. For these GRAD and CURV values, the final solution is independent from the adaptative gridding procedure. All the flame calculations were performed using multi-component transport properties and including the Soret effects. The ignition delay times were calculated based on the OH profiles assuming adiabatic and constant volume conditions behind the reflected shock wave.

3. Results and discussions

3.1. Laminar flame speeds

The experimental results obtained using the spherical bomb technique are reported in Fig. 8a for the three initial temperatures considered using the styrene from Acros Organics as fuel. The increment in the flame speed due to the variation in the initial temperature is similar for a similar increase in temperature from 342 to 373 K and from 373 to 405 K. In particular, the profiles at around the maximum differ by around 8 cm/s, while they slightly converge at fuel rich and lean conditions. Figure 8a also contains the simulation profiles. Overall the experimental results are well reproduced by the model over the entire equivalence ratio range. The maximum discrepancy between the simulations and the experiments is lower than 4%, in any case lower than the experimental uncertainties for most of the conditions considered.

The results presented in Fig. 8a were also used to estimate the influence of the initial temperature on the adiabatic laminar burning velocity based on the relationship

$$S_L/S_{Lref} = \left(T/T_{ref}\right)^{\alpha} \quad (2)$$

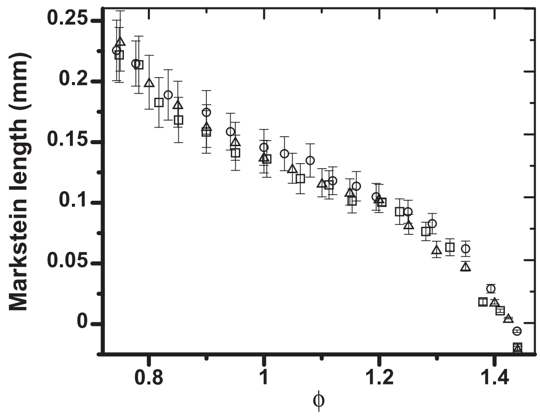

Fig. 9. Markstein lengths L′ of styrene/air mixtures, P_{in} = 100 kPa. Initial temperature equal to: circles 405 K, triangles 373 K, squares 342 K.

where the reference temperature T_{ref} is, in this case, equal to 342 K. The experimental and modeling results are presented in Fig. 8b, where the error bars were obtained by the least-squares procedure. α varies between 1.57 at $\phi = 1.15$ and 1.79 at $\phi = 0.75$, while the maximum error due to the fit in the experimental data is around 2%. The mild dependence of the experimental power exponents on the equivalence ratio is well reproduced by the model although the simulation profiles slightly underpredict the experimental ones at fuel lean conditions with a discrepancy lower than 5%. This result is in agreement with the excellent agreement between the simulations and the experimental flame speeds observed in Fig. 8a for all the initial temperature conditions.

The solution of Eq. 1 also provides the value of the Markstein length L_b and consequently of the renormalized burned Markstein length L′ ($L' = L_b/\sigma$). Since the unstretched laminar flame speed of the unburned gases, $S_u^0 = S_b^0/\sigma$, is discussed in the manuscript, L′ is presented for consistency (in fact we could re-write Eq. 1 in terms of S_u, S_u^0, and L′). The results for the three experimental sets in Fig. 8 are reported in Fig. 9. Considering the experimental errors, no influence of the initial temperature on the response of the flames to the stretch can be observed. Note also that the Markstein lengths are positive for equivalence ratios between 0.75 and 1.40, thus in almost the entire domain the flames are stable. The conclusions drawn in the work by Jayachandran et al. [23] show how the Markstein length of fuel rich mixtures is mainly influenced by the differential diffusion of the fuel compared to the oxygen. Following such analysis, the ratio between the oxygen diffusivity and the styrene diffusivity at $\phi = 1.4$ ($P = 1$ atm, $T = 298$ K) was estimated to be around 3. This value is very similar to the corresponding ratio for n-heptane, thus a similar behavior is expected. Concerning *n*-heptane, Wu et al. [19] presented some representative experimental profiles of the spatial flame velocity versus stretch rate which indicate that the Markstein length becomes negative for values of the equivalence ratio between 1.3 (positive L_b) and 1.4 (negative L_b). This is not very different from the results obtained in the present work where L_b is nearly zero at $\phi \sim 1.4$, in support of the fact that the differential diffusion may play a relevant role in the flame propagation speed as a function of κ. In reality, the analysis is more complicated since the rate at which the fuel is consumed as well as the diffusivities of the intermediate molecules should be studied and compared in detail. This analysis could be the focus of an entire, separate paper. Additional comparisons using DNS simulations would also bring light on the different aspects influencing the Markstein length of different fuel molecules.

The Markstein number Ma′ was also estimated based on the measured L′, and in this case a definition of flame thickness has to be provided. Two different definitions are generally used, one based on the temperature gradient obtained from the simulations (δ_{gradT} according to Poinsot and Veynante [44]), the other based on the thermal diffusivity ($l_f = D_{th}/S_L$). In particular, the thermal diffusivity of the unburned gases was utilized here, similarly to the work by Bechtold and Matalon [45] or the recent work by Giannakopoulos et al. [46] who compared with excellent agreement the DNS calculations on spherically expanding flames with the asymptotic theory.

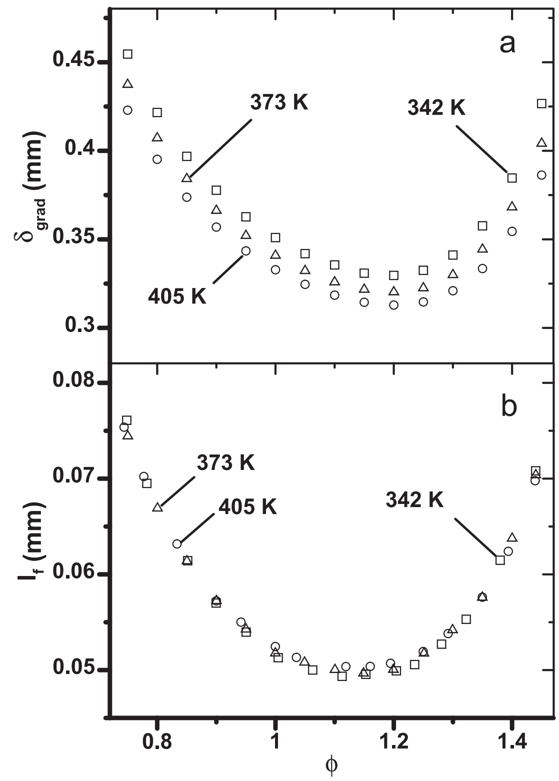

Fig. 10. Flame thickness of styrene/air flames, based on: (a) temperature gradient; (b) thermal diffusivity. P_{in} = 100 kPa, initial temperature equal to: circles 405 K, triangles 373 K, squares 342 K.

The calculated flame thicknesses of styrene/air mixtures are presented in Fig. 10. While a measurable variation of δ_{gradT} with the initial temperature is observed (the flame thickness diminishes with an increase in T_{in}), the flame thicknesses obtained based on the thermal diffusivities are similar for the three experimental sets. Moreover the difference between the two methods is substantial, varying between a factor of 5.5–6.8. This consideration poses clearly some limitations in the definition of the absolute values of the experimental Markstein numbers for which a definition of flame thickness should be given. In particular, l_f was used to derive the experimental values in Fig. 11b as for the theoretical formulation in [46].

In addition to the experimentally derived Ma′, the dimensionless Markstein number relative to the unburned gas was calculated using the asymptotic theories, in particular according to Bechtold and Matalon [45], Egolfopoulos and Law [47], and Addabbo et al. [48]. The relations solved in order to derive Ma_u are reported in the Supplementary Information, while the calculated and experimental Markstein numbers are shown in Fig. 11. As extensively discussed by Davis and Searby [49], the experimental and theoretical results cannot be directly compared from a quantitative point of view since they are associated to different Markstein lengths. On the other hand, the general trends show some differences. In particular, at fuel lean conditions (below equivalence ratio 0.9) the calculated Ma profile is nearly constant, with a subsequent

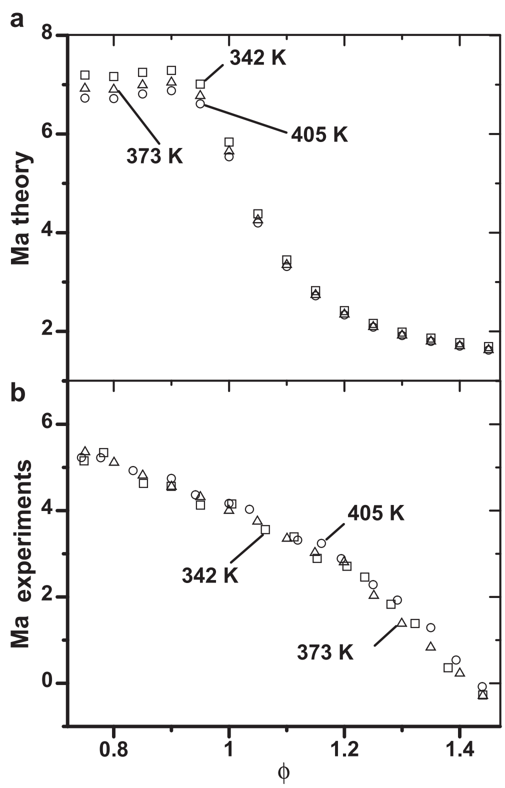

Fig. 11. Markstein numbers of styrene/air flames, based on (a) theory, associated to L_u; (b) experiments, associated to L'. P_{in} = 100 kPa, initial temperature equal to: circles 405 K, triangles 373 K, squares 342 K.

Table 2
Initial mixture compositions in mole fractions for data presented in Fig. 12.

	ϕ	Styrene	O_2	Ar
99% Argon	1.5	0.0013	0.0087	0.99
	1.0	0.0009	0.0091	0.99
	0.5	0.00048	0.00952	0.99
0.272% Styrene	1.5	0.00272	0.0181	0.97918
	1.0	0.00272	0.02719	0.97009
	0.5	0.00272	0.0544	0.94288
5% Oxygen	1.5	0.0075	0.05	0.9425
	1.0	0.005	0.05	0.945
	0.5	0.0025	0.05	0.9475

decrease and a final leveling at fuel rich conditions. On the other hand, the profile of Ma′ in Fig. 11b is nearly monotonically decrescent with a higher slope above $\phi \sim 1.2$. Finally, the results in Fig. 11b do not suggest any influence of the initial temperature on the Markstein number, while a small difference in the theoretical profiles is noticeable (with decreased Ma numbers at higher temperatures) . On the other hand, such difference is below 3.5%, thus negligible considering the uncertainties due to the assumptions implemented to derive such values.

3.2. Ignition delay times

The experimental and modeling results on the ignition delay times are presented in Fig. 12. The study was conducted varying the equivalence ratio between 0.5 and 1.5 while maintaining fixed the percentage of argon diluent at 99% (Fig. 12a), or the initial fuel mole fraction at 0.272% (Fig. 12b), or the initial oxygen mole fraction at 5% (Fig. 12c). The compositions of the initial mixtures are reported in Table 2. The pressure behind the reflected shock

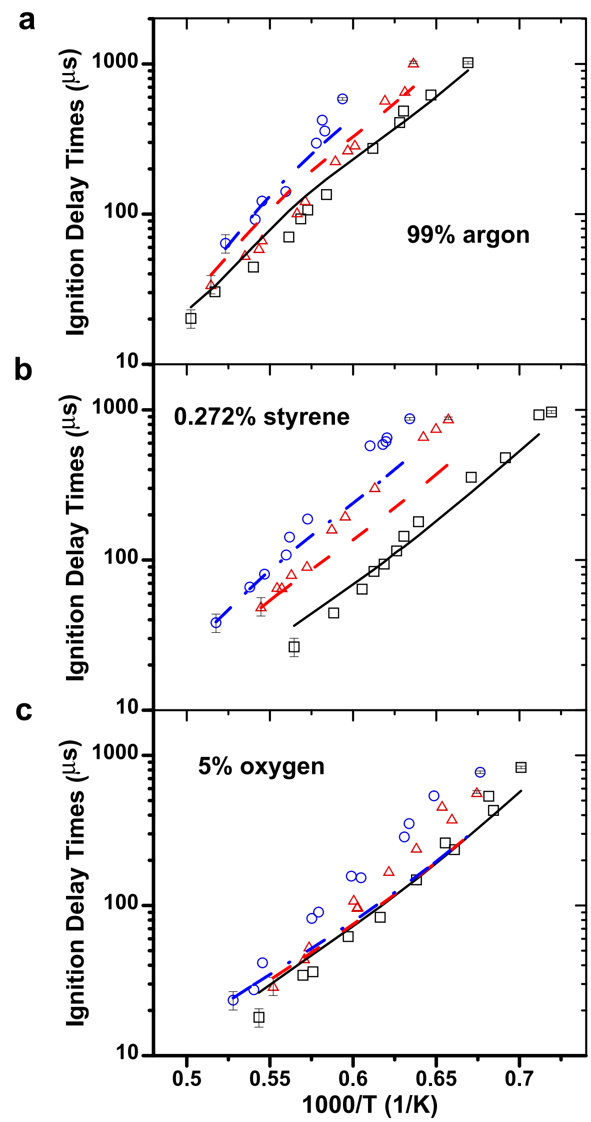

Fig. 12. Ignition delay times of styrene/O_2 mixtures in argon. Circles, – • – ϕ = 1.5, triangles, – – ϕ = 1.0, squares, —– ϕ = 0.5. Lines represent modeling results.

wave varied between 110 and 200 kPa depending on the driver gas pressure necessary to obtain the desired temperature value.

The overall correlation for the ignition delay times of styrene, based on the general expression discussed in detail by Lifshitz [50], has been derived from the experimental results as

$$\tau = A[C_8H_8]^a[O_2]^b \exp\left(\frac{E_a}{RT}\right)$$

where $A = 4.193 \times 10^{-8}$, $a = 0.848 \pm 0.040$, $b = -1.476 \pm 0.046$, $Ea/R = 24{,}344.5 \pm 387$ K, and where the concentrations are expressed in mol/cm^3, the temperatures in *K*, and the delay times in μs. The coefficients and the corresponding standard errors were obtained using a multiple linear regression (method of least squares). The comparison between the normalized ignition delay data and the correlation function obtained by statistical analysis (Fig. 13) provides a value of $r^2 \sim 0.976$. The normalized ignition delay time is here defined as the ratio between the ignition delay time and the relative concentrations of fuel, oxygen, and argon elevated at the corresponding coefficients (a, b).

Figure 12 also contains the results of the modeling simulations. The simulations are in excellent agreement with the experiments at highly diluted conditions (99% argon bath gas) although some

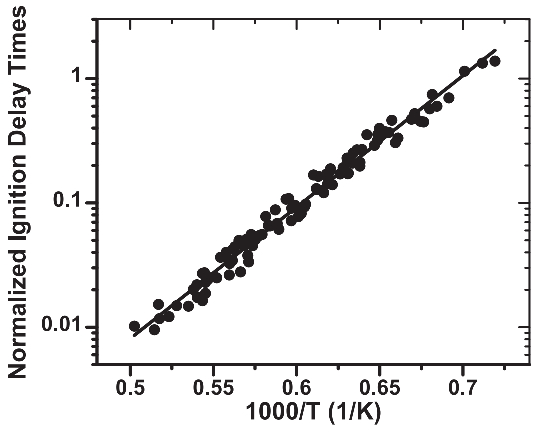

Fig. 13. Normalized ignition delay times (symbols) and correlation function (line).

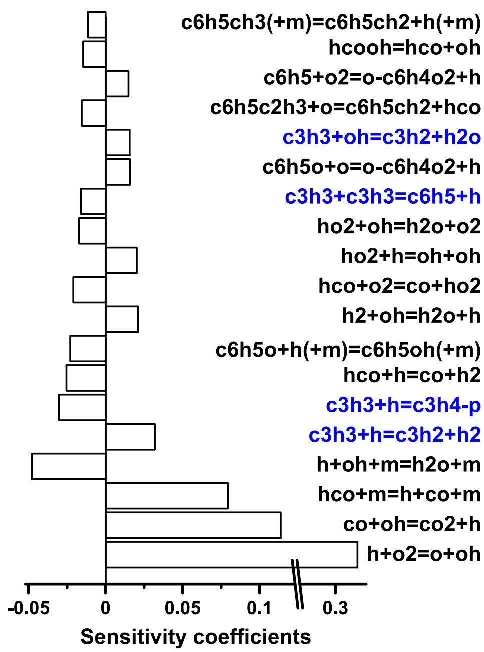

Fig. 14. Sensitivity analysis of the laminar flame speed with respect to the reaction rate constants, ϕ = 1.1, T_{in} = 342 K.

discrepancies are observed at less diluted conditions (argon mole fractions between 94.29% and 97.92% for the experiments conducted at constant fuel mole fraction, between 94.25% and 94.75% at constant oxygen mole fraction). In particular, the experimental ignition delay times below 100 μs of the stoichiometric and fuel rich mixtures presented in Fig. 12b are well predicted by the model while longer ignition delay times are underestimated. In addition, the dependence of the ignition delay times on the fuel concentration (Fig. 12c) is not correctly reproduced (the modeling profiles nearly overlay).

3.3. Chemical kinetic analyses

The chemical kinetic model accurately simulates the flame speed measurements as shown in Fig. 8, thus it can be used to further analyze the styrene oxidation at the present flame conditions. In particular, Fig. 14 contains the sensitivity analysis of the flame speed with respect to the reaction rate constants for an equivalence ratio of 1.1 and an initial temperature equal to 342 K. Of course the reactions of the H_2/O_2 sub-model, the oxidation of CO by OH and the HCO reactions play the most relevant role. In

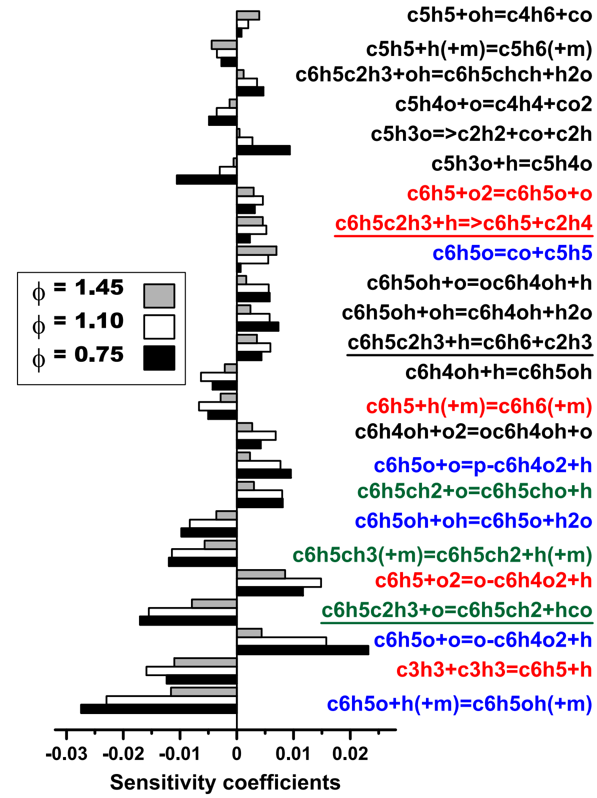

Fig. 15. Sensitivity analysis of the laminar flame speed with respect to the reaction rate constants of reactions involving only the aromatic chemistry. T_{in} = 342 K.

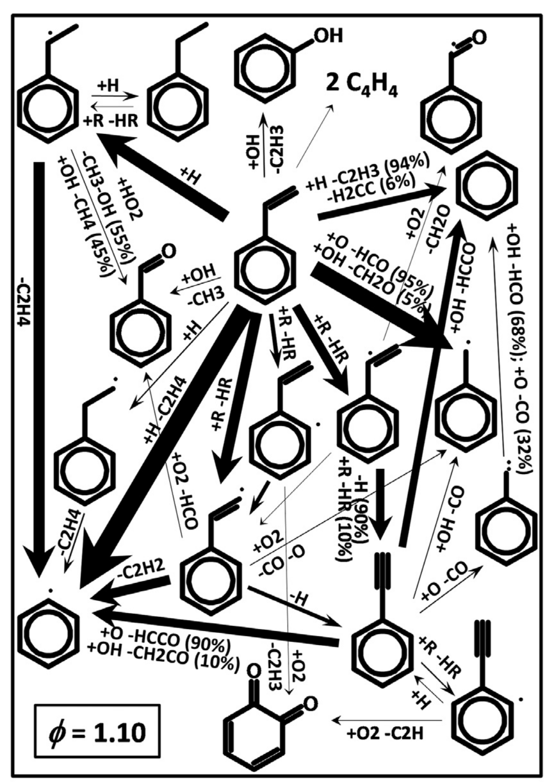

Fig. 16. Rate of production, ϕ = 1.10, P = 100 kPa, and T_{in} = 342 K. Arrow thickness proportional to the integrated reaction rates normalized based on the initial fuel mole fraction compared to the stoichiometric condition.

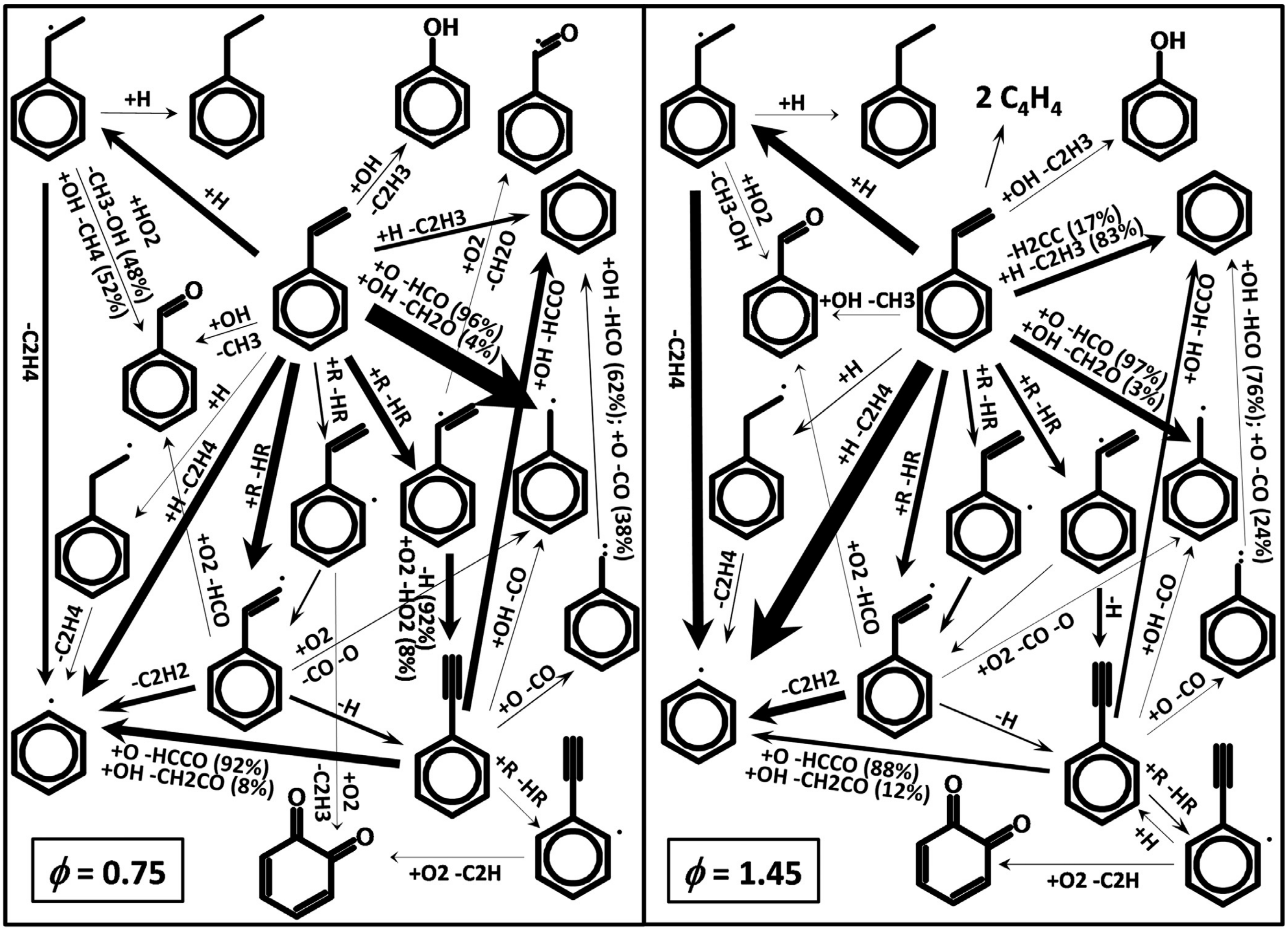

Fig. 17. Rate of production, $\phi = 0.75$ and $\phi = 1.45$, $P = 100$ kPa, and $T_{in} = 342$ K. Arrow thickness proportional to the integrated reaction rates normalized based on the initial fuel mole fraction compared to the stoichiometric condition.

addition, the chemistry of the propargyl radical strongly influences the propagation of the styrene/air flames (reactions in blue). The propargyl radical is formed mainly by the reaction of the phenyl radical with atomic hydrogen (inverse of the reaction $2C_3H_3 = C_6H_5+H$). This reaction accounts for nearly 91% of the total C_3H_3 formation (based on rate of production analysis), while a small contribution from the oxidation of vinylacetylene is also present (~4%).

The sensitivity analysis in Fig. 14 contains also some specific reactions involving the aromatic ring and the fuel. In order to better understand the influence of the reactions specifically investigated here (thus concerning the styrene chemistry) on the flame speed, and how these reactions couple with the aromatic chemistry by Metcalfe et al. [30], a second analysis was performed considering only the reactions involving the aromatic chemistry, at similar conditions of equivalence ratio (1.1) and initial temperature (342 K) (Fig. 15, white bars). The results clearly indicate how the flame speed of styrene is influenced by the chemistry of the single ring radicals, such as the phenoxy radical (C_6H_5O, blue reactions), the phenyl radical (C_6H_5, red reactions), and the benzyl radical ($C_6H_5CH_2$, green reactions), with competition between the oxidation steps and the recombination reactions to form stable intermediates. For example, the reaction of the phenoxy radical (C_6H_5O) with hydrogen to form phenol (C_6H_5OH) has the greatest coefficient, opposite in sign to the coefficients for the displacement reactions with molecular oxygen to form the benzoquinones

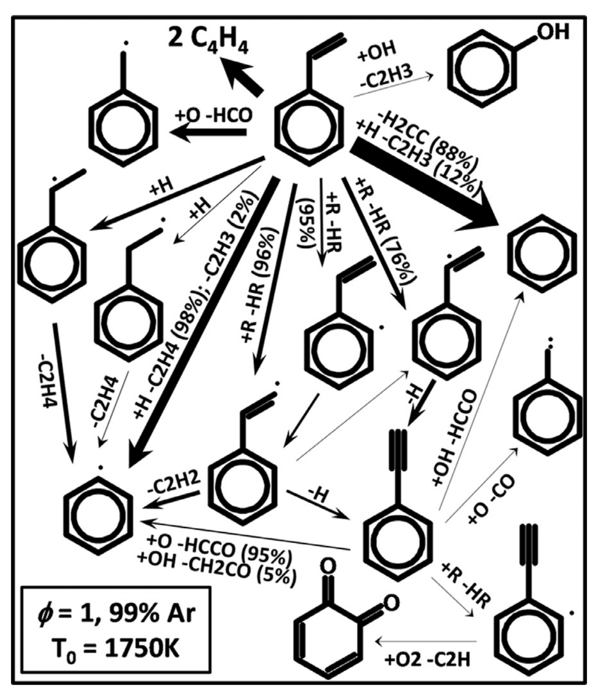

Fig. 18. Rate of production analysis at 50% fuel decay, $\phi = 1$, $T_0 = 1750$ K, $P_0 = 148$ kPa, 99% argon.

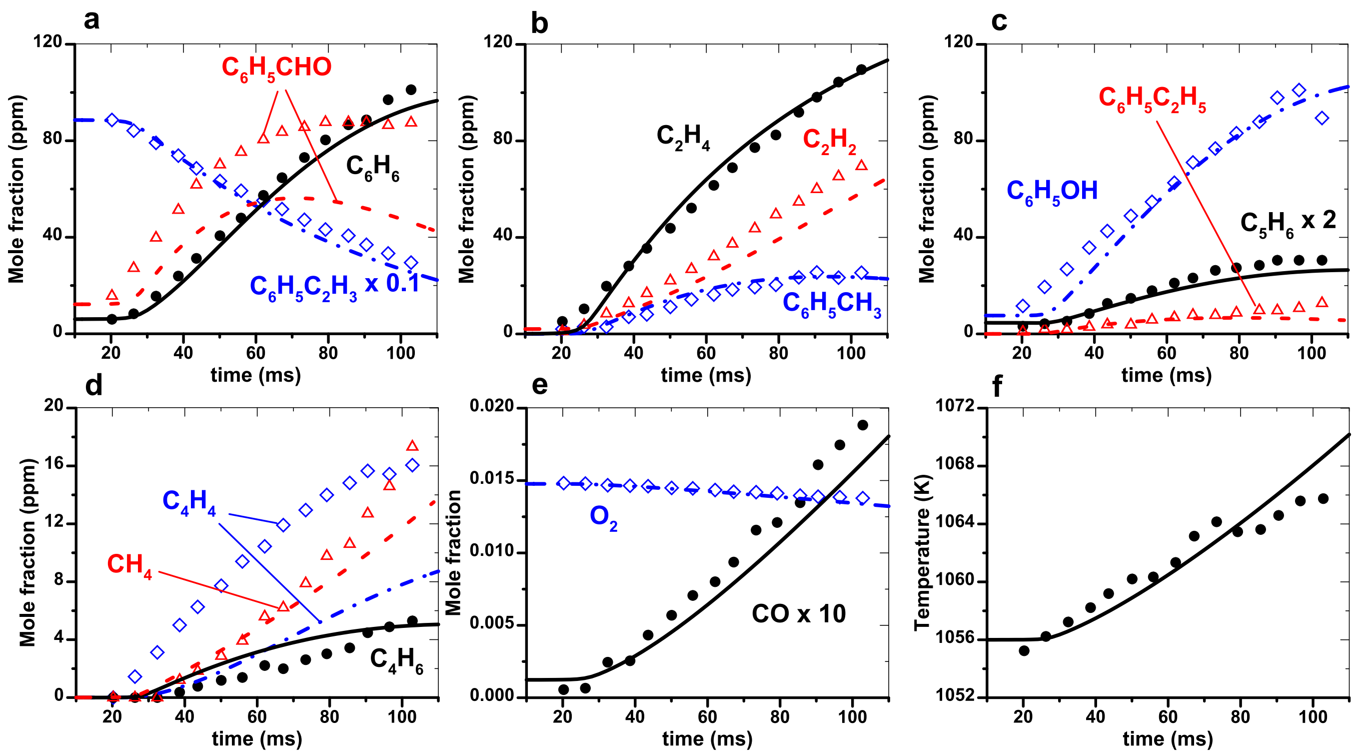

Fig. 19. Atmospheric plug flow reactor data, styrene/O_2 mixture in N_2, ϕ = 0.56 (symbols, Litzinger et al. [1]) and simulations (lines, present model). Experimental data shifted by 12 ms.

(p-$C_6H_4O_2$ and o-$C_6H_4O_2$) and for the decomposition step to CO and cyclopentadienyl radical (C_5H_5). Similarly, the oxidation steps of the phenyl radical (C_6H_5) and the benzyl radical ($C_6H_5CH_2$) are in competition with the reactions with H atoms to benzene (C_6H_6) or propargyl radicals and to toluene ($C_6H_5CH_3$), respectively.

The sensitivity analysis was repeated also at different equivalence ratios to investigate the relative importance of the different reactions. At lean conditions (black bars), among the reactions specific to the aromatic chemistry the laminar flame speed is mostly influenced by the phenoxy radical reactions as well as by the reactions involving the C_5H_3O intermediate (produced from the oxidation of the ring) which do not play a relevant role at fuel rich conditions. This result confirms the prominent role of the oxidation steps at these conditions. On the contrary, at ϕ = 1.45 the sensitivity coefficients of the two main phenoxy radical reactions become smaller in absolute value, while the thermal decomposition step to cyclopentadienyl radical (C_5H_5) and CO becomes more relevant. As a consequence, the chemistry of the cyclopentadienyl radical influences the flame speed, with the competition between the oxidation step (C_5H_5+OH = C_4H_6+CO) and the recombination with H to form cyclopentadiene.

Among the reactions specific to the fuel molecule, three main reactions of styrene with radicals appear in Fig. 15 (underlined reactions). In particular, the formation of benzyl radical and formyl radical from the reaction between styrene and atomic oxygen reduces the flame speed mainly due to the formation of the very stable benzyl radicals which act as H-atom scavengers. On the other hand, the reactions of the fuel with hydrogen atoms to form the phenyl radical + ethylene and benzene + vinyl radical lead to the formation of very reactive C2 compounds, thus they contribute positively to an increase in the flame speed.

These reactions also constitute three of the major fuel consumption pathways, as shown in the rate of production analysis presented in Fig. 16 (ϕ = 1.10 and T_{in} = 342 K). Around 23%, 20% and 9% of the integrated flux from the fuel leads directly to the formation of phenyl, benzyl, and benzene, respectively, through the above mentioned reactions, with small contributions from the thermal decomposition of styrene to benzene + vinylidene and the reaction of styrene with OH to form benzyl + formaldehyde. In addition, the H-abstraction reactions to form 2-phenylvinyl, 1-phenylvinyl, and 2-vinylphenyl contribute to around 29% of the fuel consumption. 2-phenylvinyl mainly decomposes to C_6H_5 and C_2H_2 (64%) and to phenylacetylene + H (27%), while 1-phenylvinyl to phenylacetylene + H. Finally, around 15% of the fuel undergoes H-addition to form $C_6H_5CHCH_3$ which quickly decomposes to phenyl and ethylene.

The rate of production analysis was also performed at fuel-rich and fuel-lean conditions (Fig. 17). At ϕ = 1.45 the reactions of the fuel with hydrogen clearly become the major pathways for the fuel consumption, with increased contributions from the thermal decomposition of the fuel to benzene + vinylidene and through the opening of the ring to form vinylacetylene molecules (C_4H_4), as also observed by Yuan et al. [12] for rich flames. On the contrary, at fuel-lean conditions the predominant pathways are the reaction of styrene with molecular oxygen to form benzyl (31%) and the H-abstraction reactions (37%).

In order to better understand the relevance of the different reaction pathways for the fuel consumption at the highly diluted conditions encountered in the shock tube study compared to the flame case, rate of production analysis are presented in Fig. 18 for an experiment conducted at stoichiometric conditions, T_0 = 1750 K and P_0 =148 kPa, and dilution equal to 99%. At these high temperature conditions, the fragmentation of the fuel becomes the predominant consumption pathway leading to the formation of benzene and vinylidene (26% of the total flux) or vinylacetylene (17%). The thermal fragmentation of the fuel with the loss of an H atom to form 2-phenylvinyl, 1-phenylvinyl, and 2-vinylphenyl complements the H-abstraction reactions which account totally

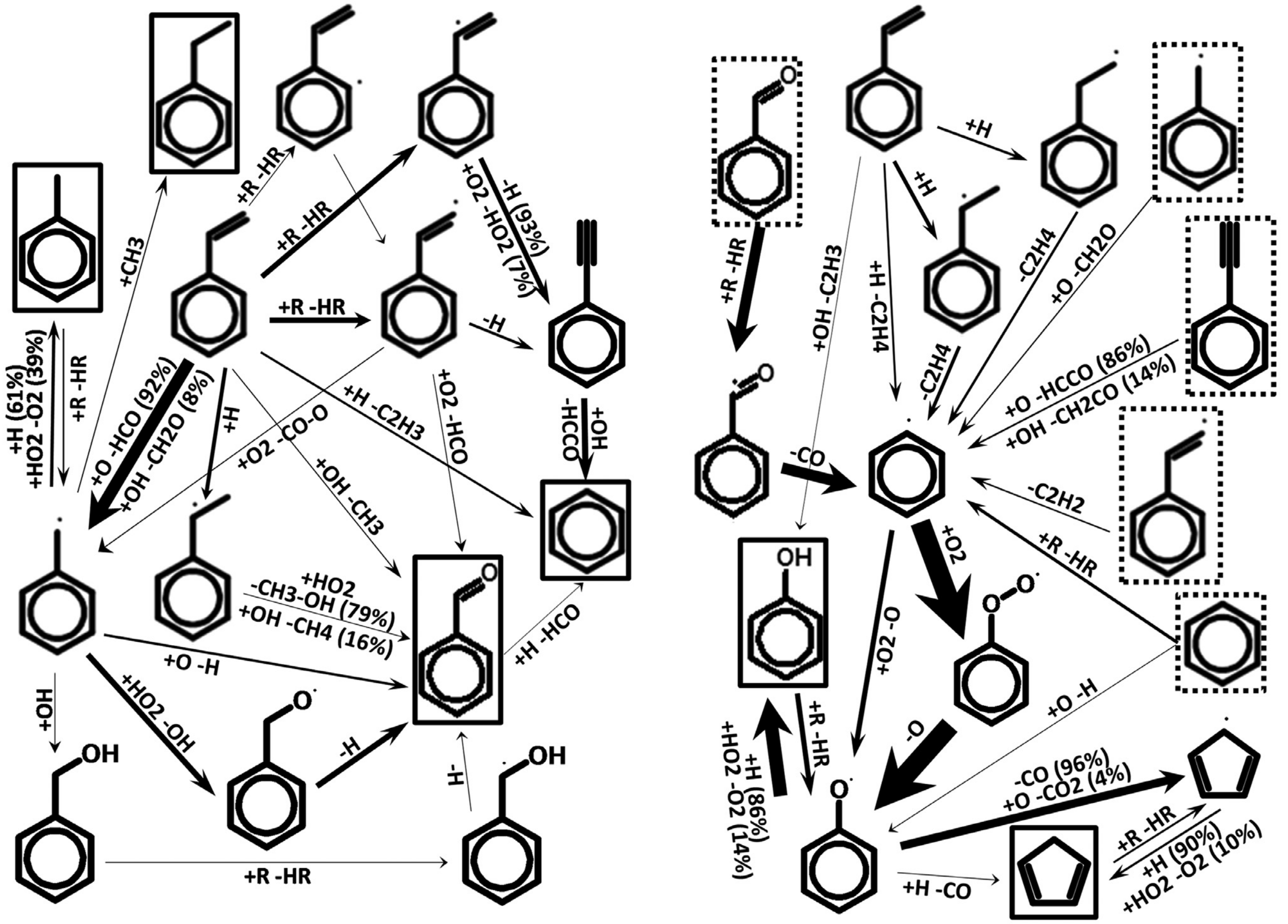

Fig. 20. Rate of production analysis at 70 ms, atmospheric plug flow reactor, $\phi = 0.56$, $T_{in} = 1056$ K.

for the 19% of the fuel consumption rate, while the reaction of styrene with O to form benzyl and HCO and with H to phenyl and C_2H_4 are still major pathways influencing the fuel consumption and consequently the autoignition process.

3.4. Model validation against species profiles

The kinetic model was also validated against the flow reactor species profiles by Litzinger et al. [1] (Fig. 19). The experiments were conducted at atmospheric pressure, for an initial mixture composed of 0.09% styrene + 1.6 % oxygen in nitrogen ($\phi \sim 0.56$), and initial temperature equal to 1056 K. Both the fuel consumption and the formation of the main intermediate compounds are well reproduced by the model. In particular, among the aromatic compounds only the formation of benzaldehyde is slightly underpredicted. In order to investigate the different pathways which lead to the intermediates presented in Fig. 19, the rate of production analysis was performed, and the results at a time equal to 70 ms (which corresponds to ~50% of the fuel decay) are reported in Fig. 20. For clarity of discussion, the graph was divided into two complementary sections. The species in solid-line boxes are the target intermediates modeled in Fig. 19, while for the ones in dash-line boxes (right section) the formation pathways from the fuel molecule have been omitted from the graph (they can be found in the left side section). The main fuel consumption step involves the reaction with atomic oxygen to form the benzyl radical (31% of the total fuel consumption) which is the building block for the formation of toluene through recombination with H

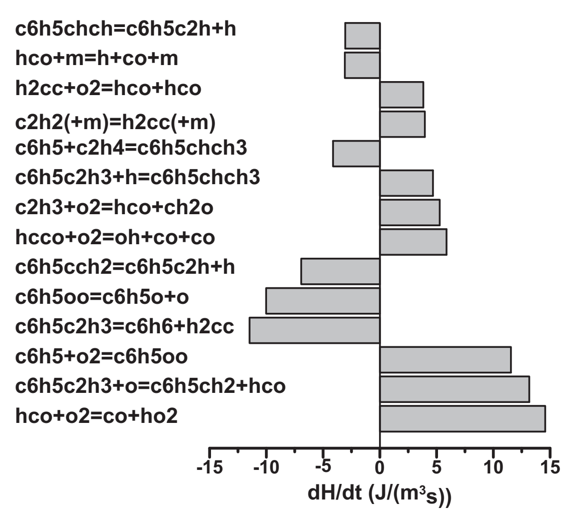

Fig. 21. Rate of heat production per reaction (J/(m^3s)). Simulations of the atmospheric plug flow reactor, styrene/O_2 mixture in N_2, $\phi = 0.56$, $t = 42$ ms.

atoms and through reaction with hydroperoxyl radicals (Fig. 19b) and for the formation of ethylbenzene through recombination with CH_3 radicals (Fig. 19c). On the other hand, benzene derives mainly from the H-abstraction reactions on the styrene ring and vinyl moiety, which result in the formation of phenylacetylene first and

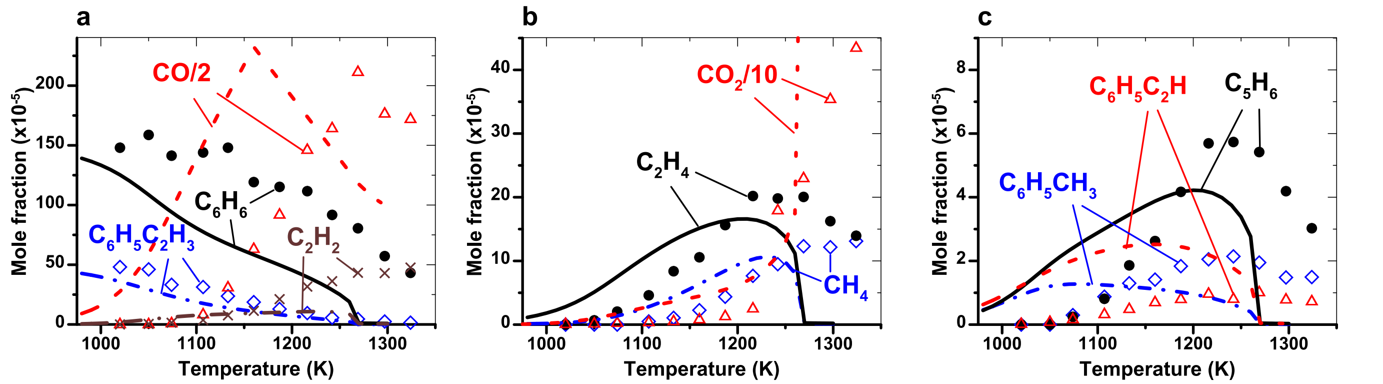

Fig. 22. Jet stirred reactor data, $\phi = 1$, $P = 1$ atm, $t = 0.07$ s (symbols, Yuan et al. [12]) and simulations (lines, present model).

C_6H_6 second through the reaction $C_6H_5C_2H$+OH = C_6H_6+HCCO. Benzene, whose profile is presented in Fig. 19a, is also formed by the ipso-addition by H to the fuel molecule (which accounts for 22% of C_6H_6 formation) and, in minor extent, by the reaction of benzaldehyde with atomic hydrogen (only 5%). The pathways for the formation of benzaldehyde are multiple, thus it is not evident the reason of the discrepancy between experiments and simulations in Fig. 19a. The main pathway proceeds through formation of benzyl radical and $C_6H_5CH_2O$ (50% of the total C_6H_5CHO formation). Other reactions involving the benzyl radical + O (20%) and styrene + OH (9%) strongly contribute to the benzaldehyde profile.

Figure 20 contains also the reaction pathways for the formation of phenol and cyclopentadiene (right scheme) whose profiles are presented in Fig. 19c. In this case the phenyl radical plays the central role. In fact its oxidation leads to the formation of the phenoxy radical which undergoes recombination with H atoms and reaction with HO_2 to form phenol (96% of the total C_6H_5OH formation, with an additional small contribution from the ipso-addition by OH to the fuel). On the other hand, the thermal fragmentation of the phenoxy radical as well as its reaction with atomic oxygen (only minor pathway) leads to the formation of the cyclopentadienyl radical. This resonantly stabilized intermediate subsequently reacts with H and HO_2 to form C_5H_6. Another small pathway to cyclopentadiene involves the reaction between the phenoxy radical and H.

Due to its central role, the formation of the phenyl radical is relevant in order to accurately simulate the profiles in Fig. 19. Nearly 41% of the C_6H_5 production derives from the thermal fragmentation of the benzoyl radical which is formed by H-abstraction reactions of benzaldehyde. Other pathways include the H-addition/C_2H_4 elimination from styrene (21% of the phenyl radical formation), the H-abstraction reactions on the benzene ring (12%), the reaction of styrene with H to form C_6H_5+C_2H_4 (7%), the decomposition of C_6H_5CHCH (6%), and the reactions of benzyl radical and phenylacetylene with molecular oxygen (5% each).

Concerning the small fragmentation intermediates, the formation of C_2H_4 is well reproduced by the model (Fig. 19b). Ethylene is mainly produced by the H-addition/C_2H_4 elimination from styrene (67% of the total rate of production) and the reaction of styrene with H forming C_6H_5 + ethylene (23%). On the other hand, butadiene (C_4H_6), whose experimental profile is compared in Fig. 19d to the simulation profile of the 1,3-butadiene isomer based on the results obtained by Yuan et al. [12] in the jet-stirred reactor and in the flame, derives from the oxidation of the cyclopentadienyl radical while acetylene (Fig. 19b) from the oxidation of the ring (in particular from the C_5H_3O intermediate, 58% of the total rate of production) and from the decomposition of 2-phenylvinyl into C_6H_5 + C_2H_2 (29%). Finally, the results in Fig. 19d show that the experimental and numerical data for vinylacetylene do not well agree (the experimental mole fractions are

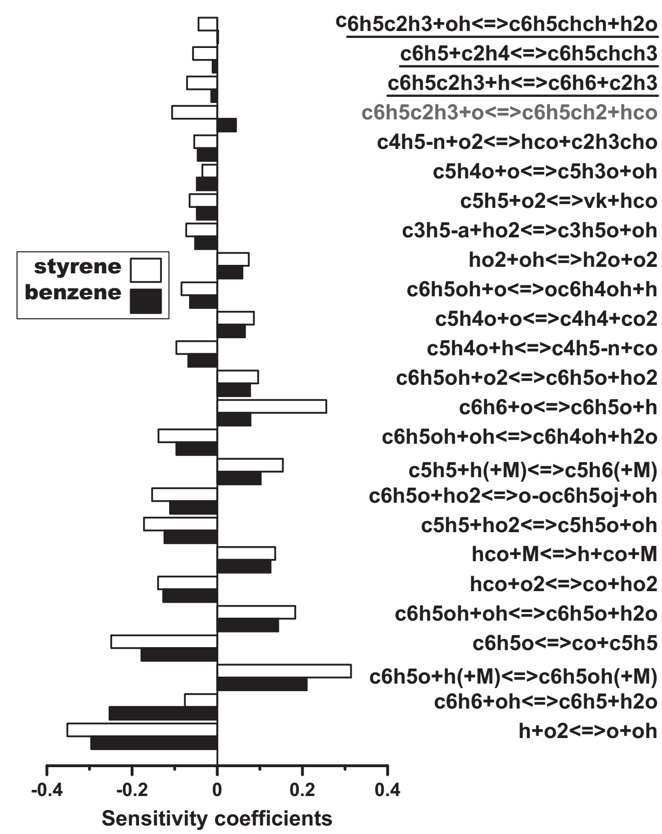

Fig. 23. Sensitivity analysis of styrene and benzene mole fractions with respect to the reaction rate constants, conditions as in Fig. 22, T = 1070 K.

substantially underpredicted by the model). In the current kinetic mechanism, C_4H_4 is produced by the oxidation of the ring and the subsequent intermediate compounds (such as C_5H_4O). The thermal decomposition of the benzyl radical to form vinylacetylene + the propargyl radical as suggested by Braun-Unkhoff et al. [51,52] is also included in the original chemical kinetic model but it doesn't contribute significantly to the formation of C_4H_4 at the flow reactor conditions. The discrepancy between model and experiments suggests the possibility of additional pathways forming C_4H_4.

The comparison presented in Fig. 19f shows how the model is also able to accurately simulate the increment in the temperature profile which follows the progress with time of the reaction system. In order to better understand the contributions of the different reactions to the temperature increase, the rates of heat production per reaction at 42 ms (corresponding to a calculated temperature ~1058 K) are presented in Fig. 21. As expected the

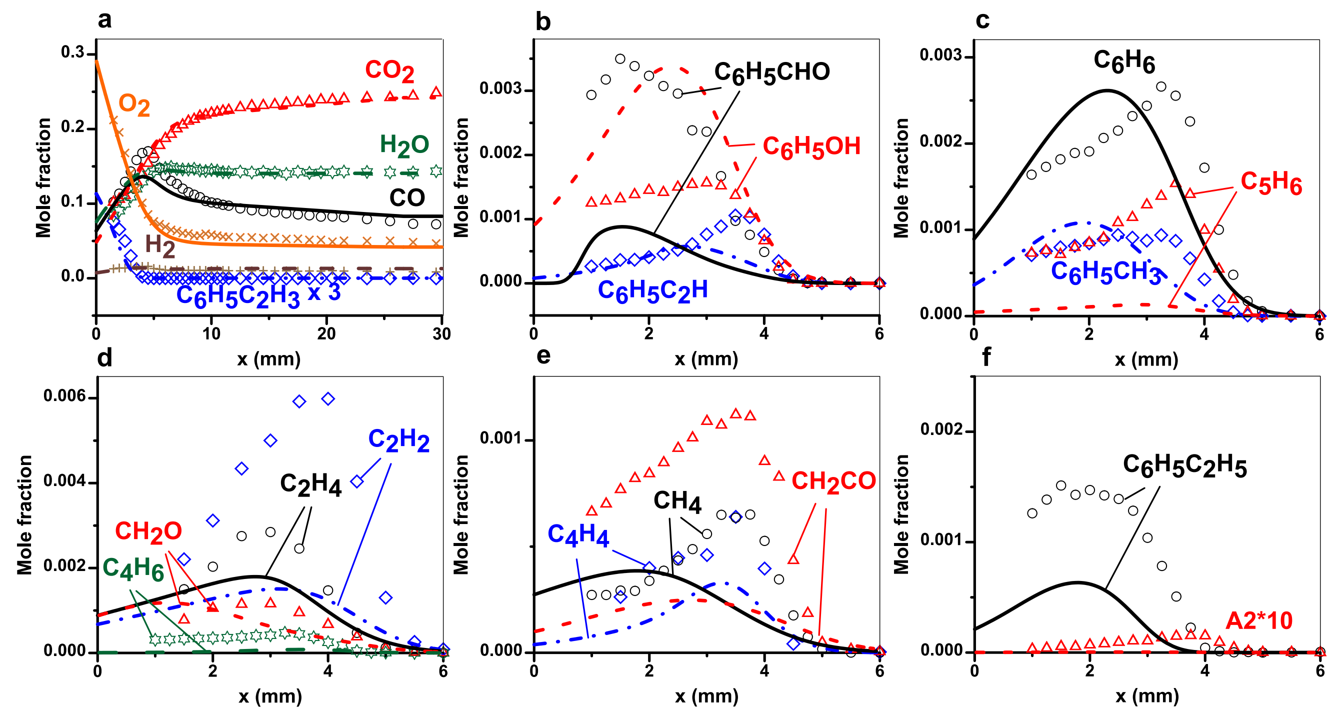

Fig. 24. Laminar premixed flame data, $\phi = 1$, $P = 0.0395$ atm (symbols, Yuan et al. [12]) and simulations (lines, present model).

formation of CO by HCO oxidation is associated with higher energy release, together with the reaction of styrene with molecular O to form the benzyl radical + HCO. The oxidation of the phenyl radical through the subsequent steps $C_6H_5+O_2 \rightarrow C_6H_5OO \rightarrow C_6H_5O+O$ globally does not result in a significant heat production rate, while most of the energy is consumed by the breaking of the fuel to form benzene and vinylidene. Other reactions which contribute to the increase in temperature include the oxidation of HCCO and C_2H_3, the H-addition to styrene to form $C_6H_5CHCH_3$ and the reactions involving vinylidene (isomerization to acetylene and oxidation).

The experimental results by Yuan et al. [12] were not used during the development of the model; nevertheless the kinetic scheme was tested against such data. In particular, the jet-stirred reactor data were obtained at three different equivalence ratios ranging between 0.5 and 1.5, with constant initial fuel mole fraction (0.05% styrene +0.15% benzene) in N_2, for a fixed residence time (0.07 s), at $P = 1$ atm and temperatures between 1000 and 1357 K. Typical profiles for the jet-stirred reactor data (styrene/benzene mixtures) at stoichiometric conditions are presented in Fig. 22. The model is not able to accurately predict the species profiles especially with respect to the oxidation and consumption of the fuel molecules. The simulations are substantially faster than the experiments, in particular for the consumption of benzene (Fig. 22a). As a consequence, the formation and consumption of the intermediate compounds is shifted towards lower temperatures (Fig. 22a, b and c). The first step for the improvement of the model is the analysis of the consumption paths of styrene and benzene, specifically how the reactions parameters influence the simulation profiles. Therefore, the sensitivity analysis of the benzene and styrene mole fractions with respect to the reaction rate coefficients was performed. The results obtained at a temperature of 1070 K (corresponding to 50% of the styrene decay) are presented in Fig. 23. The reactions not underlined have been ordered based on the coefficients for benzene (black bars); the mole fraction of this compound is predominantly influenced by the branching reaction $H+O_2 = O+OH$, by chemistry of HCO, and by the aromatic chemistry for the ring oxidation as in the original model by Metcalfe et al. [30]. The only significant reaction which is relative to the styrene chemistry is the reaction between styrene and molecular oxygen to form benzyl radical and HCO (reaction in blue), but the corresponding coefficient is relatively small, thus an adjustment of the reaction rate constant would not significantly improve the simulation results. In addition, the specific coefficients for benzene (black bar) and styrene (white bar) have opposite signs, which means that an adjustment in the rate constant aimed at improving the benzene simulations would worsen the styrene simulations, and vice versa (both mole fractions need indeed to increase at the same time).

The reactions affecting the styrene profile in Fig. 22 are similar to the ones for benzene. In addition, other reactions specific to the sub-model developed in the present investigation appear in the analysis (underlined reactions in Fig. 23), but the corresponding coefficients are relatively small. The consumption of styrene is then mainly driven by the fact that benzene is present in the initial fuel mixture and the corresponding chemistry is too fast compared to the experimental trends. Thus, the base benzene chemistry used in the present work affects the prediction capability with respect to the jet stirred reactor data on styrene/benzene mixtures presented in [12]. In order to improve the simulation results, modifications in the reactions from the Metcalfe et al. [30] model would be necessary, which is beyond the scope of this study, also considering the fact that the original model was extensively validated against a large number of experimental results relative to the oxidation and pyrolysis of cyclopentadiene, benzene, and toluene. In this case, the choice not to modify any reaction in the original model of course limits our capability to improve the simulation results obtained in the jet-stirred reactor.

Finally, Fig. 24 shows the simulations of the laminar flame mole fraction profiles obtained by Yuan et al. [12]. As for the jet-stirred reactor case, only the stoichiometric results are reported here as an example, although the flame data in Ref. [12] were obtained at different equivalence ratios (0.75, 1.00, and 1.70) but at the same pressure (0.0395 atm), argon percentage (50%), and inlet velocity (35 cm/s). The profiles of styrene, oxygen, and the main combustion products are well reproduced by the model

(Fig. 24a). Concerning the aromatic compounds formed during the consumption and oxidation of the fuel molecule, the general trends are captured but discrepancies can be observed for some of the profiles. In particular, the formation of phenol is overpredicted by the model, in contrast with the benzaldehyde profile which is underpredicted. The mole fractions of phenylacetylene (Fig. 24b), benzene, and toluene (Fig. 24c) are quite well simulated, while surprisingly the formation of cyclopentadiene is not well reproduced. This is possibly due to the chemistry of the phenoxy radical as suggested in Ref. [12]. The experimental and simulation results for other intermediates are presented in Fig. 24d, e, and f. In general the predicted drop in the intermediates profiles starts earlier in the flame compared to the experiments, thus a possible reason for the observed discrepancies is the high reactivity of the intermediate compounds. On the other hand, the experimental results were obtained at very low pressure (0.0395 atm) while no pressure dependency was considered for the reaction rate parameters of the styrene sub-model studied in the present work. For example, the rate constant parameters of the reactions involving the decomposition and isomerization of C_6H_5CHCH and $C_6H_5CCH_2$, now from the transition state theory calculations by Tokmakov and Lin [36], or the recombination of the benzyl radical with CH_3 to form ethylbenzene will need to be revised including Troe parameters to account for the variation with pressure. In addition, the original toluene sub-mechanism should be adjusted to better predict the toluene low-pressure flames by Li et al. [53]. In particular, the acetylene and cyclopentadienyl formation are underpredicted by the model as shown in the Supplementary Information for the stoichiometric mixture. This is possibly linked to the underpredictions observed in Fig. 24c and d for C_2H_2 and C_5H_6. Once again, the pressure dependency of the ring oxidation steps should be analyzed in detail. Other investigations will be necessary to further improve the capability of the current model in relation to the prediction of the very low-pressure chemistry.

4. Conclusions

Laminar flame speeds of styrene/air mixtures were measured for the first time in a spherical bomb heated to 342 K, 373 K, and 405 K at an initial pressure of 100 kPa and over a wide range of equivalence ratios. In addition, ignition delay time measurements were performed in a glass shock tube over a wide range of equivalence ratios and dilutions in argon bath gas, for initial pressures between 110 and 200 kPa. The shock tube experiments are also the first measurements obtained on the ignition of styrene as a fuel. The experimental results were used to develop a chemical kinetic model which is able to accurately simulate the measured flame speeds as well as the temperature dependence observed in the results, in addition to the ignition delay times and the flow reactor data available in literature. The model was subsequently used to perform sensitivity and reaction pathway analyses to clarify the relevance of the various reactions at the different experimental conditions considered herein. The comparison between the simulation and experimental results on low-pressure and jet-stirred reactor data indicates the need for further improvement in the kinetic model in relation to the corresponding experimental conditions.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2016.08.026.

References

[1] T.A. Litzinger, K. Brezinsky, I. Glassman, The oxidation of ethylbenzene near 1060 K, Combust. Flame 63 (1986) 251–267.
[2] T.A. Litzinger, K. Brezinsky, I. Glassman, Reactions of N-Propylbenzene During Gas Phase Oxidation, Combust. Sci. Technol. 50 (1986) 117–133.
[3] K. Brezinsky, The high-temperature oxidation of aromatic hydrocarbons, Prog. Energy Combust. Sci. 12 (1986) 1–24.
[4] K. Brezinsky, G.T. Linteris, T.A. Litzinger, I. Glassman, High temperature oxidation of n-alkyl benzenes, Symp. (Int.) Combust. 21 (1988) 833–840.
[5] P. Dagaut, A. Ristori, A. El Bakali, M. Cathonnet, Experimental and kinetic modeling study of the oxidation of n-propylbenzene, Fuel 81 (2002) 173–184.
[6] P. Diévart, P. Dagaut, The oxidation of n-butylbenzene: Experimental study in a JSR at 10 atm and detailed chemical kinetic modeling, Proc. Combust. Inst. 33 (2011) 209–216.
[7] B. Husson, R. Bounaceur, K. Tanaka, M. Ferrari, O. Herbinet, P.A. Glaude, R. Fournet, F. Battin-Leclerc, M. Crochet, G. Vanhove, R. Minetti, C.J. Tobin, K. Yasunaga, J.M. Simmie, H.J. Curran, T. Niass, O. Mathieu, S.S. Ahmed, Experimental and modeling study of the oxidation of n-butylbenzene, Combust. Flame 159 (2012) 1399–1416.
[8] S. Gudiyella, K. Brezinsky, High pressure study of n-propylbenzene oxidation, Combust. Flame 159 (2012) 940–958.
[9] J.D. Bittner, J.B. Howard, Composition profiles and reaction mechanisms in a near-sooting premixed benzene/oxygen/argon flame, Symp. (Int.) Combust. 18 (1981) 1105–1116.
[10] M. Frenklach, D.W. Clary, W.C. Gardiner Jr, S.E. Stein, Detailed kinetic modeling of soot formation in shock-tube pyrolysis of acetylene, Symp. (Int.) Combust. 20 (1985) 887–901.
[11] N.W. Moriarty, N.J. Brown, M. Frenklach, Hydrogen Migration in the Phenylethen-2-yl Radical, J. Phys. Chem. A 103 (1999) 7127–7135.
[12] W. Yuan, Y. Li, P. Dagaut, J. Yang, F. Qi, Experimental and kinetic modeling study of styrene combustion, Combust. Flame 162 (2015) 1868–1883.
[13] W. Müller-Markgraf, J. Troe, Thermal decomposition of ethylbenzene, styrene, and bromophenylethane: UV absorption study in shock waves, J. Phys. Chem. 92 (1988) 4914–4922.
[14] M.A. Grela, V.T. Amorebieta, A.J. Colussi, Pyrolysis of styrene: kinetics and mechanism of the equilibrium styrene .tautm. benzene + acetylene, J. Phys. Chem. 96 (1992) 9861–9865.
[15] D. Nativel, M. Pelucchi, A. Frassoldati, A. Comandini, A. Cuoci, E. Ranzi, N. Chaumeix, T. Faravelli, Laminar flame speeds of pentanol isomers: an experimental and modeling study, Combust. Flame 166 (2016) 1–18.
[16] P.D. Ronney, G.I. Sivashinsky, A theoretical study of propagation and extinction of nonsteady spherical flame fronts, SIAM J. Appl. Math. 49 (1989) 1029–1046.
[17] A.P. Kelley, C.K. Law, Nonlinear effects in the extraction of laminar flame speeds from expanding spherical flames, Combust. Flame 156 (2009) 1844–1851.
[18] COSILAB, The combustion simulation laboratory, Rotexo GmbH & Co., KG, Haan, Germany, 2009 Version 3.3.2. <http://www.SoftPredict.com>..
[19] F. Wu, W. Liang, Z. Chen, Y. Ju, C.K. Law, Uncertainty in stretch extrapolation of laminar flame speed from expanding spherical flames, Proc. Combust. Inst. 35 (2015) 663–670.
[20] P. Clavin, Dynamic behaviour of premixed flame fronts in laminar and turbulent flows, Prog. Energy Comb. Sci. 11 (1985) 1–59.
[21] M. Matalon, B.J. Matkowsky, Flames as gasdynamic discontinuities, J. Fluid Mech. 124 (1982) 239–259.
[22] Z. Chen, On the extraction of laminar flame speed and Markstein length from outwardly propagating spherical flames, Combust. Flame 158 (2011) 291–300.
[23] J. Jayachandran, R. Zhao, F.N. Egolfopoulos, Determination of laminar flame speeds using stagnation and spherically expanding flames: Molecular transport and radiation effects, Combust. Flame 161 (2014) 2305–2316.
[24] J. Santner, F.M. Haas, Y. Ju, F.L. Dryer, Uncertainties in interpretation of high pressure spherical flame propagation rates due to thermal radiation, Combust. Flame 161 (2014) 147–153.
[25] Y. Ju, G. Masuya, P. Ronney, Effect of radiative emission and absorption on the propagation and extinction of premixed gas flames, Symp. (Int.) Combust. 27 (1998) 2619–2626.
[26] Z. Chen, Effects of radiation and compression on propagating spherical flames of methane/air mixtures near the lean flammability limit, Combust. Flame 157 (2010) 2267–2276.
[27] H. Yu, W. Han, J. Santner, X. Gou, C.H. Sohn, Y. Ju, Z. Chen, Radiation-induced uncertainty in laminar flame speed measured from propagating spherical flames, Combust. Flame 161 (2014) 2815–2824.
[28] M. Yahyaoui, N. Djebaïli-Chaumeix, C.E. Paillard, S. Touchard, R. Fournet, P.A. Glaude, F. Battin-Leclerc, Experimental and modeling study of 1-hexene oxidation behind reflected shock waves, Proc. Combust. Inst. 30 (2005) 1137–1145.
[29] K.S. Khuong, W.H. Jones, W.A. Pryor, K.N. Houk, The mechanism of the self--Initiated thermal polymerization of styrene. Theoretical solution of a classic problem, J. Am. Chem. Soc. 127 (2005) 1265–1277.
[30] W.K. Metcalfe, S. Dooley, F.L. Dryer, Comprehensive detailed chemical kinetic modeling study of toluene oxidation, Energy Fuels 25 (2011) 4915–4936.
[31] V.D. Knyazev, A. Bencsura, S.I. Stoliarov, I.R. Slagle, Kinetics of the $C_2H_3 + H_2 = H + C_2H_4$ and $CH_3 + H_2 = H + CH_4$ reactions, J. Phys. Chem. 100 (1996) 11346–11354.

[32] H. Wang, E. Dames, B. Sirjean, D.A. Sheen, R. Tangko, A. Violi, J.Y.W. Lai, F.N. Egolfopoulos, D.F. Davidson, R.K. Hanson, C.T. Bowman, C.K. Law, W. Tsang, N.P. Cernansky, D.L. Miller, R.P. Lindstedt, A high-temperature chemical kinetic model of n-alkane (up to n-dodecane), cyclohexane, and methyl-, ethyl-, n-propyl and n-butyl-cyclohexane oxidation at high temperatures, JetSurF version 2.0, September 19, 2010.
[33] K. Mahmud, P. Marshall, A. Fontijn, A high-temperature photochemistry kinetics study of the reaction of oxygen(3P) atoms with ethylene from 290 to 1510 K, J. Phys. Chem. 91 (1987) 1568–1573.
[34] W. Tsang, Chemical kinetic data base for combustion chemistry part V. Propene, J. Phys. Chem. Ref. Data 20 (1991) 221–273.
[35] C. Saggese, A. Frassoldati, A. Cuoci, T. Faravelli, E. Ranzi, A wide range kinetic modeling study of pyrolysis and oxidation of benzene, Combust. Flame 160 (2013) 1168–1190.
[36] I.V. Tokmakov, M.C. Lin, Reaction of phenyl radicals with acetylene: quantum chemical investigation of the mechanism and master equation analysis of the kinetics, J. Am. Chem. Soc. 125 (2003) 11397–11408.
[37] I.V. Tokmakov, M.C. Lin, Combined quantum chemical/RRKM-ME computational study of the phenyl + ethylene, vinyl + benzene, and H + styrene reactions, J. Phys. Chem. A 108 (2004) 9697–9714.
[38] A. Fahr, S.E. Stein, Reactions of vinyl and phenyl radicals with ethyne, ethene and benzene, Symp. (Int.) Combust. 22 (1989) 1023–1029.
[39] T. Seta, M. Nakajima, A. Miyoshi, High-temperature reactions of OH radicals with benzene and toluene, J. Phys. Chem. A 110 (2006) 5081–5090.
[40] K. Narayanaswamy, G. Blanquart, H. Pitsch, A consistent chemical mechanism for oxidation of substituted aromatic species, Combust. Flame 157 (2010) 1879–1898.
[41] M. Eichholtz, A. Schneider, D.V. Stucken, J.T. Vollmer, H.G. Wagner, The reactions of phenylacetylenes with $O(^3P)$ in the gas phase, Symp. (Int.) Combust. 25 (1994) 859–866.
[42] H. Richter, S. Granata, W.H. Green, J.B. Howard, Detailed modeling of PAH and soot formation in a laminar premixed benzene/oxygen/argon low-pressure flame, Proc. Combust. Inst. 30 (2005) 1397–1405.
[43] H. Nakamura, D. Darcy, M. Mehl, C.J. Tobin, W.K. Metcalfe, W.J. Pitz, C.K. Westbrook, H.J. Curran, An experimental and modeling study of shock tube and rapid compression machine ignition of n-butylbenzene/air mixtures, Combust. Flame 161 (2013) 49–64.
[44] T. Poinsot, D. Veynante, Theoretical and Numerical Combustion, in: R.T. Edwards (Ed.), Flourtown, 2005.
[45] J.K. Bechtold, M. Matalon, The dependence of the Markstein length on stoichiometry, Combust. Flame 127 (2001) 1906–1913.
[46] G.K. Giannakopoulos, A. Gatzoulis, C.E. Frouzakis, M. Matalon, A.G. Tomboulides, Consistent definitions of “Flame Displacement Speed” and “Markstein Length” for premixed flame propagation, Combust. Flame 162 (2015) 1249–1264.
[47] F.N. Egolfopoulos, C.K. Law, Chain mechanisms in the overall reaction orders in laminar flame propagation, Combust. Flame 80 (1990) 7–16.
[48] F.R. Addabbo, J.K. Bechtold, M. Matalon, Wrinkling of spherically expanding, Proc. Combust. Inst. 29 (2002) 1527–1535.
[49] S.G. Davis, G. Searby, The use of counterflow flames for the evaluation of burning velocities and stretch effects in hydrogen/air mixtures, Combust. Sci. Technol. 174 (2002) 93–110.
[50] A. Lifshitz, “Ignition Delay Times,” Chapter 16.5 in Handbook of Shock Waves Volume 3, G. Ben-Dor, O. Igra, T. Elperin and A. Lifshitz, Eds. Academic Press, San Diego (2001).
[51] M. Braun-Unkhoff, P. Frank, T. Just, A shock tube study on the thermal decomposition of toluene and of the phenyl radical at high temperatures, Symp. (Int.) Combust. 22 (1989) 1053–1061.
[52] M. Braun-Unkhoff, P. Frank, T. Just, High temperature reactions of benzyl radicals, Ber. Buns. Phys. Chem. 94 (1990) 1417–1425.
[53] Y. Li, J. Cai,L. Zhang, T. Yuan, K. Zhang, F. Qi, Investigation on chemical structures of premixed toluene flames at low pressure, Proc. Combust. Inst. 33 (2011) 593–600.